

STAT 201A: Introduction to Probability at an Advanced Level

Lecture Notes (in development)

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Note: These draft lecture notes are a work in progress. They may contain typos, mistakes, and inaccuracies.

These notes accompany the course *STAT 201A: Introduction to Probability at an Advanced Level*. They aim to present the foundations of probability theory highlighting connections to statistics, machine learning, and other applications.

• Bayes formula
• Monty Hall
• Discrete r.v.

Lecture 0 and Lecture 1

We start with some basic examples. If A and B are events with $\mathbb{P}(B) > 0$, then the conditional probability of A given B is defined as

$$\mathbb{P}(A | B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$

A useful situation is when we have a *complete system of events*, say $A_1, \dots, A_k$. This means that exactly one of these events happens: they are disjoint and their union is the whole sample space.

Using this system we can derive the **Bayes formula**. Suppose B is another event with $\mathbb{P}(B) > 0$. We start with the definition:

$$\mathbb{P}(A_j | B) = \frac{\mathbb{P}(A_j \cap B)}{\mathbb{P}(B)}.$$

Next, note that

$$\mathbb{P}(A_j \cap B) = \mathbb{P}(B | A_j) \mathbb{P}(A_j).$$

Also, since $\{A_1, \dots, A_k\}$ is a complete system of events, we can write

$$\mathbb{P}(B) = \sum_{i=1}^k \mathbb{P}(B | A_i) \mathbb{P}(A_i).$$

Putting everything together gives

$$\mathbb{P}(A_j | B) = \frac{\mathbb{P}(B | A_j) \mathbb{P}(A_j)}{\sum_{i=1}^k \mathbb{P}(B | A_i) \mathbb{P}(A_i)}.$$

This is the Bayes formula, which expresses how our belief in A_j changes once we observe the event B .

Problem 1 (Monty Hall). You are on a game show with three doors. One hides a car and two hide goats. You pick Door 1. The host, who knows the locations, opens Door 3 showing a goat, never opens your chosen door, always opens a goat, and—if both remaining doors have goats—chooses uniformly at random. He offers you Door 2. Is it to your advantage to switch?

Example 1 (Monty Hall). Fix the initial choice to be Door 1. Let

$$C_i = \{\text{car behind door } i\}, \quad i = 1, 2, 3, \quad H_j = \{\text{host opens door } j\}, \quad j = 1, 2, 3.$$

We observed H_3 . By Bayes,

$$\mathbb{P}(C_2 | H_3) = \frac{\mathbb{P}(H_3 | C_2) \mathbb{P}(C_2)}{\sum_{i=1}^3 \mathbb{P}(H_3 | C_i) \mathbb{P}(C_i)}.$$

With “priors” $\mathbb{P}(C_i) = \frac{1}{3}$ and host behavior

$$\mathbb{P}(H_3 | C_2) = 1, \quad \mathbb{P}(H_3 | C_1) = \frac{1}{2}, \quad \mathbb{P}(H_3 | C_3) = 0,$$

we get

$$\mathbb{P}(C_2 | H_3) = \frac{1 \cdot \frac{1}{3}}{\frac{1}{2} \cdot \frac{1}{3} + 1 \cdot \frac{1}{3} + 0 \cdot \frac{1}{3}} = \frac{2}{3}.$$

Thus after Door 3 is opened (goat), switching to Door 2 wins with probability $2/3$, while staying with Door 1 wins with probability $1/3$; switching is advantageous.

It appears that the strategy of the host for this problem is really important. Let us discuss another example.

Problem 2 (Monty Hall with a random host). There are three doors; one hides a car, two hide goats. You pick Door 1. The host does *not* remember where the car is and opens uniformly at random one of the two other doors. He might reveal the car; if so, you may choose it. If he reveals a goat (say, Door 3), you are offered to switch to the remaining closed door (Door 2). What strategy maximizes the win probability?

Example 2 (Monty Hall with a random host). Let $C_i = \{\text{car behind door } i\}$ and $H_j = \{\text{host opens door } j\}$. Priors: $\mathbb{P}(C_i) = 1/3$. Host rule: given your initial Door 1, choose uniformly from $\{2, 3\}$, regardless of the car's location.

$$\begin{aligned} \mathbb{P}(G_3, H_3 | C_1) &= \mathbb{P}(H_3 | C_1, G_3) \cdot \mathbb{P}(G_3 | C_1) = \frac{1}{2} \times 1 = \frac{1}{2} \\ \mathbb{P}(G_3, H_3 | C_2) &= \mathbb{P}(H_3 | C_2, G_3) \cdot \mathbb{P}(G_3 | C_2) = \frac{1}{2} \times 1 = \frac{1}{2} \\ \mathbb{P}(C_1 | \text{goat}, H_3) &= \frac{\mathbb{P}(G_3, H_3 | C_1) \cdot \mathbb{P}(C_1)}{\mathbb{P}(G_3, H_3 | C_1) \cdot \mathbb{P}(C_1) + \mathbb{P}(G_3, H_3 | C_2) \cdot \mathbb{P}(C_2) + \mathbb{P}(G_3, H_3 | C_3) \cdot \mathbb{P}(C_3)} = 0 \end{aligned}$$

Case 1: host opens a goat (e.g., H_3 and we see a goat).

$$\mathbb{P}(H_3 | C_2) = \frac{1}{2}, \quad \mathbb{P}(H_3 | C_1) = \frac{1}{2}, \quad \mathbb{P}(H_3 | C_3) = 0 \text{ (would reveal the car).}$$

Bayes gives

$$\mathbb{P}(C_2 | H_3, \text{goat}) = \frac{(\frac{1}{2})(\frac{1}{3})}{(\frac{1}{2})(\frac{1}{3}) + (\frac{1}{2})(\frac{1}{3})} = \frac{1}{2}, \quad \mathbb{P}(C_1 | H_3, \text{goat}) = \frac{1}{2}.$$

So after a random goat reveal, staying vs. switching are equivalent (both $1/2$).

Case 2: host opens the car (say, Door 2). Then choosing the revealed car is optimal (win with probability 1).

Reminder: Discrete random variables and expectation. A random variable (r.v.) X is a real-valued quantity whose exact value is uncertain to us. For “reminders only”, we focus here on *discrete* X (finite or countable support). We will go through all of this in more detail later.

A discrete X has a support $S_X = \{a_1, a_2, \dots\}$ and a probability mass function (pmf)

$$p_X(a) = \mathbb{P}(X = a), \quad p_X(a) \geq 0, \quad \sum_{a \in S_X} p_X(a) = 1.$$

We sometimes write $p(a)$ instead of $p_X(a)$. For any function $g : \mathbf{R} \rightarrow \mathbf{R}$, the expectation is defined as follows

$$\mathbb{E}[g(X)] = \sum_{a \in S_X} g(a) p_X(a),$$

whenever the series converges absolutely. In particular,

$$\mathbb{E}[X] = \sum_{a \in S_X} a p_X(a).$$

Basic properties of expectation:

- *Linearity:* $\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y]$ (no independence needed).
- *Constants:* $\mathbb{E}[c] = c$.
- *Indicator trick:* for an event A , $\mathbb{E}[\mathbf{1}(A)] = \mathbb{P}(A)$. If $X = \sum_k \mathbf{1}(A_k)$ counts how many A_k occur, then $\mathbb{E}[X] = \sum_k \mathbb{P}(A_k)$.

Problem 3 (Six-legged landing on a mostly-land planet). A spherical planet has land region L on its surface with area fraction $a = \text{area}(L) \in (0, 1)$ (normalize total surface area to 1). Assume $a > 1/2$. A spacecraft has six fixed leg tips at points $p_1, \dots, p_6$ on a unit sphere attached to the craft. When attempting to land, the planet may be oriented by any rotation; under a rotation R the tips would touch the planet at $Rp_1, \dots, Rp_6$. A landing is called *successful* if at least four tips touch land. Show that a successful landing orientation exists.



Example 3 (Landing is possible). Choose a rotation of the planet uniformly at random (rotation-invariant choice). For $i = 1, \dots, 6$ define the indicator random variable

$$X_i = \mathbf{1}(Rp_i \in L).$$

Let $S = X_1 + \dots + X_6$ be the number of legs that land on land.

Step 1: Compute $\mathbb{E}[X_i]$. By symmetry (uniform rotation), the point Rp_i is uniformly distributed on the sphere, so

$$\mathbb{E}[X_i] = \mathbb{P}(Rp_i \in L) = \text{area}(L) = a > \frac{1}{2}.$$

Step 2: Linearity of expectation. Independence is *not* assumed (and in fact the X_i are typically dependent), but linearity gives

$$\mathbb{E}[S] = \sum_{i=1}^6 \mathbb{E}[X_i] = 6a > 3.$$

Assume $P(X \leq 3) = 1$,
 $\Rightarrow P(X=0) + P(X=1) + \dots + P(X=3) = 1$
 $\mathbb{E}[X] = 0 \cdot P(X=0) + \dots + 3 \cdot P(X=3)$
 $\leq 3 \cdot [P(X=0) + \dots + P(X=3)]$
 $= 3$

Step 3: Existence. Since $S \in \{0, 1, \dots, 6\}$ is integer-valued, the inequality $\mathbb{E}[S] > 3$ implies $\mathbb{P}(S \geq 4) > 0$; otherwise, if $S \leq 3$, then $\mathbb{E}[S] \leq 3$, a contradiction. Hence there exists a rotation with $S \geq 4$, i.e., a successful landing.

Example 4 (Bribes in Monty Hall: standard and forgetful host). Fix your initial choice to be Door 1. Let

$$C_i = \{\text{car behind door } i\}, \quad i = 1, 2, 3, \quad H_j = \{\text{host opens door } j\}, \quad j = 1, 2, 3.$$

Let the car value be $v > 0$. A bribe $b \geq 0$ is paid iff you “stay” (commit to not switch) after the host opens a door.

We formalize payoffs with random variables (defined after we see which door is opened):

$$X_{\text{st}} = b + v \mathbf{1}(C_1), \quad X_{\text{sw}} = v \mathbf{1}(C_2) \quad \text{when the host opens Door 3;}$$

if instead the host opens Door 2, swap the roles of 2 and 3 in the definition of X_{sw} . We compare conditional expectations (in our case defined on the conditional probability space) given the observed event.

(a) *Standard host (knows location, never opens your door, always opens a goat).* If the host opens Door 3, then it must show a goat. Using your earlier calculation,

$$\mathbb{P}(C_1 | H_3) = \frac{1}{3}, \quad \mathbb{P}(C_2 | H_3) = \frac{2}{3}.$$

Hence

$$\mathbb{E}[X_{\text{st}} | H_3] = b + v \mathbb{P}(C_1 | H_3) = b + \frac{v}{3}, \quad \mathbb{E}[X_{\text{sw}} | H_3] = v \mathbb{P}(C_2 | H_3) = \frac{2v}{3}.$$

Accept the bribe (stay) iff

$$b + \frac{v}{3} \geq \frac{2v}{3} \iff b \geq \frac{v}{3}.$$

Otherwise, decline and switch.

(b) *Forgetful (random) host (chooses uniformly among the two unchosen doors)*. Introduce

$G_3 = \{\text{host opens Door 3 and shows a goat}\}$, $K_2 = \{\text{host opens Door 2 and shows the car}\}$.

Goat revealed (observe G_3). Then

$$\mathbb{P}(C_1 | G_3) = \mathbb{P}(C_2 | G_3) = \frac{1}{2},$$

so with the same $X_{\text{st}}, X_{\text{sw}}$ as above,

$$\mathbb{E}[X_{\text{st}} | G_3] = b + \frac{v}{2}, \quad \mathbb{E}[X_{\text{sw}} | G_3] = \frac{v}{2}.$$

Accept the bribe iff $b \geq 0$.

Car revealed (observe K_2). Then C_2 holds surely, so staying yields

$$\mathbb{E}[X_{\text{st}} | K_2] = b \quad (\text{since } \mathbf{1}(C_1) = 0),$$

and switching gives the car with certainty:

$$\mathbb{E}[X_{\text{sw}} | K_2] = v.$$

Accept the bribe iff $b \geq v$; otherwise switch to the revealed car.

Example 5 (Group testing). We have N individuals; each is infected independently with probability p (small). Partition into groups of size k (assume $(k | N)$). For each group, mix the k samples and run one pooled test. If the pooled test is negative, all k are cleared. If the pooled test is positive, test the k individuals separately. The aim is to choose k to minimize the expected number of tests per person (to save cost).

Single group: expected tests. Let Y be the indicator r.v. that the *first group's* pooled test is positive ($Y = 1$ if positive, $Y = 0$ otherwise). By symmetry, it suffices to analyze a single group. Then the number of tests for that group is

$$T_{\text{grp}} = 1 + kY.$$

The pooled test is negative exactly when all k are uninfected, with probability $(1 - p)^k$. Hence $\mathbb{P}(Y = 1) = 1 - (1 - p)^k$ and, by the indicator trick,

$$\mathbb{E}[T_{\text{grp}}] = 1 + k\mathbb{P}(Y = 1) = 1 + k(1 - (1 - p)^k) = 1 + k - k(1 - p)^k.$$

Expected tests per person are

$$f(k) = \frac{\mathbb{E}[T_{\text{grp}}]}{k} = \frac{1}{k} + 1 - (1 - p)^k.$$

Small- p optimization. For small p and moderate k , $(1-p)^k \approx 1 - pk$. Then

$$f(k) \approx \frac{1}{k} + pk.$$

Minimizing over real $k > 0$ gives $k^* \approx 1/\sqrt{p}$ and

$$f(k^*) \approx \frac{1}{k^*} + pk^* \approx 2\sqrt{p}.$$

This computation can be made numerically *arbitrarily* precise for general p and k .

1 Lecture 2

Rules of probability and inclusion-exclusion formula

- $\mathbb{P}(A) \in [0, 1]$,
- $\mathbb{P}(A \cap B) = \mathbb{P}(A|B)\mathbb{P}(B) = \mathbb{P}(B|A)\mathbb{P}(A)$
- If the events A, B are disjoint $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B)$.
- $\mathbb{P}(\emptyset) = 0$.
- $\mathbb{P}(\bar{A}) = 1 - \mathbb{P}(A)$, where A is the complementary event.
- Monotonicity: if $A \subset B$, then $\mathbb{P}(A) \leq \mathbb{P}(B)$.

Inclusion-exclusion formula for two events. For any two events:

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B).$$

In general, for events $A_1, \dots, A_n$, write $AB = A \cap B$. Define

$$S_k = \sum_{1 \leq i_1 < \dots < i_k \leq n} \mathbb{P}(A_{i_1} \dots A_{i_k}), \quad k = 1, \dots, n.$$

Then

$$\mathbb{P}\left(\bigcup_{i=1}^n A_i\right) = \sum_{k=1}^n (-1)^{k+1} S_k = \sum_i \mathbb{P}(A_i) - \sum_{i < j} \mathbb{P}(A_i A_j) + \sum_{i < j < \ell} \mathbb{P}(A_i A_j A_\ell) - \dots$$

Problem 4 (Hat-check). N visitors check indistinguishable hats. At the end, hats are returned in a uniformly random permutation. What is the probability that *at least one* visitor gets their own hat back?

N 个人, N 个帽子, 随机打乱分配, 求至少 1 个人拿到属于自己的帽子的概率:

令 $A_i = \{i\text{-th person gets his hat}\}$

则 $P = \Pr\left(\bigcup_{i=1}^N A_i\right)$

$$= \sum_{k=1}^N (-1)^{k+1} S_k \quad \text{where } S_k = \sum_{1 \leq i_1 < i_2 < \dots < i_k \leq N} \mathbb{P}(A_{i_1} A_{i_2} \dots A_{i_k})$$

$$\begin{aligned}
& \cdot P(A_{i_1} \cdots A_{i_k}) = \frac{(N-k)!}{N!} \\
& \cdot S_k = \binom{N}{k} \cdot \frac{(N-k)!}{N!} = \frac{N!}{k!(N-k)!} \cdot \frac{(N-k)!}{N!} = \frac{1}{k!} \\
& \Rightarrow P = \sum_{k=1}^N (-1)^{k+1} \cdot S_k = \sum_{k=1}^N (-1)^{k+1} \cdot \frac{1}{k!}
\end{aligned}$$

注: 考虑 $n \rightarrow \infty$ 时的情况:
 注意到 $e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!}$,
 因此 $-e^{-1} = \sum_{k=0}^{\infty} (-1)^k \cdot \frac{1}{k!} = 1 + \sum_{k=1}^{\infty} (-1)^{k+1} \cdot \frac{1}{k!}$

Example 6 (Inclusion-exclusion for fixed points of a random permutation). Let A_i be the event that visitor i receives their own hat ($i = 1, \dots, N$). We want $\mathbb{P}(\bigcup_{i=1}^N A_i)$.

By inclusion-exclusion,

因此 $P \rightarrow 1 - e^{-1}$ as $N \rightarrow \infty$

$$\mathbb{P}\left(\bigcup_{i=1}^N A_i\right) = \sum_{k=1}^N (-1)^{k+1} \sum_{1 \leq i_1 < \dots < i_k \leq N} \mathbb{P}(A_{i_1} \cdots A_{i_k}).$$

Fix any k distinct indices $\{i_1, \dots, i_k\}$. The event $A_{i_1} \cdots A_{i_k}$ says those k visitors get their own hats; the remaining $N - k$ hats are permuted arbitrarily among the remaining $N - k$ visitors. Hence

$$\mathbb{P}(A_{i_1} \cdots A_{i_k}) = \frac{(N-k)!}{N!}.$$

There are $\binom{N}{k}$ such k -tuples, so the k -th term equals

$$(-1)^{k+1} \binom{N}{k} \frac{(N-k)!}{N!} = (-1)^{k+1} \frac{1}{k!}.$$

Therefore

$$\mathbb{P}\left(\bigcup_{i=1}^N A_i\right) = \sum_{k=1}^N (-1)^{k+1} \frac{1}{k!} = 1 - \frac{1}{2!} + \frac{1}{3!} - \dots + (-1)^{N+1} \frac{1}{N!}.$$

As $N \rightarrow \infty$,

$$\sum_{k=1}^N (-1)^{k+1} \frac{1}{k!} \rightarrow 1 - e^{-1}.$$

Let us give another simple example using the same inequalities.

Problem 5 (Three-set inequalities). Let A, B, C be events. Prove:

- (a) $\mathbb{P}(AB) + \mathbb{P}(AC) + \mathbb{P}(BC) \geq \mathbb{P}(A) + \mathbb{P}(B) + \mathbb{P}(C) - 1$;
- (b) $\mathbb{P}(AB) + \mathbb{P}(AC) - \mathbb{P}(BC) \leq \mathbb{P}(A)$.

Solution: (a) By inclusion-exclusion for three events,

$$\mathbb{P}(A \cup B \cup C) = \mathbb{P}(A) + \mathbb{P}(B) + \mathbb{P}(C) - \mathbb{P}(AB) - \mathbb{P}(BC) - \mathbb{P}(AC) + \mathbb{P}(ABC).$$

Since $\mathbb{P}(A \cup B \cup C) \leq 1$ and $\mathbb{P}(ABC) \geq 0$, we obtain

$$1 \geq \mathbb{P}(A) + \mathbb{P}(B) + \mathbb{P}(C) - \mathbb{P}(AB) - \mathbb{P}(BC) - \mathbb{P}(AC),$$

which rearranges to

$$\mathbb{P}(AB) + \mathbb{P}(AC) + \mathbb{P}(BC) \geq \mathbb{P}(A) + \mathbb{P}(B) + \mathbb{P}(C) - 1.$$



(b) Apply inclusion–exclusion to $A(B \cup C)$:

$$\mathbb{P}(A(B \cup C)) = \mathbb{P}(AB) + \mathbb{P}(AC) - \mathbb{P}(ABC).$$

Note that $A(B \cup C) \subseteq A$ and $ABC \subseteq BC$, hence by monotonicity

$$\mathbb{P}(A(B \cup C)) \leq \mathbb{P}(A) \quad \text{and} \quad -\mathbb{P}(ABC) \geq -\mathbb{P}(BC).$$

Therefore

$$\mathbb{P}(AB) + \mathbb{P}(AC) - \mathbb{P}(BC) \leq \mathbb{P}(AB) + \mathbb{P}(AC) - \mathbb{P}(ABC) = \mathbb{P}(A(B \cup C)) \leq \mathbb{P}(A),$$

which proves

$$\mathbb{P}(AB) + \mathbb{P}(AC) - \mathbb{P}(BC) \leq \mathbb{P}(A).$$

Independence of events and random variables

Definition 1 (Independent events). Events $A_1, \dots, A_k$ are independent if for every nonempty $S \subseteq \{1, \dots, k\}$,

$$\mathbb{P}\left(\bigcap_{i \in S} A_i\right) = \prod_{i \in S} \mathbb{P}(A_i).$$

Definition 2 (Independent random variables). Random variables $X_1, \dots, X_k$ are independent if for all sets $B_1, \dots, B_k \subset \mathbf{R}$,

$$\mathbb{P}\left(\bigcap_{i=1}^k \{X_i \in B_i\}\right) = \prod_{i=1}^k \mathbb{P}(X_i \in B_i).$$

This “for all sets” should be read as “for all Borel sets” in a fully formal treatment. We avoid this definition in this class. For our purposes, an equivalent formulation is

$$\mathbb{P}(X_1 \leq t_1, \dots, X_k \leq t_k) = \prod_{i=1}^k \mathbb{P}(X_i \leq t_i) \quad \text{for all } t_1, \dots, t_k \in \mathbf{R}.$$

2 Lecture 3

Definition 3 (Pairwise independence). $X_1, \dots, X_k$ are pairwise independent if for all $i \neq j$ and Borel sets $B_i, B_j \subset \mathbf{R}$,

$$\mathbb{P}(X_i \in B_i, X_j \in B_j) = \mathbb{P}(X_i \in B_i) \mathbb{P}(X_j \in B_j).$$

Pairwise independence does not imply independence of the whole collection.

Example 7 (Pairwise independent but not jointly independent). Let U, V be independent Rademacher variables: these are the random signs $\mathbb{P}(U = \pm 1) = \mathbb{P}(V = \pm 1) = \frac{1}{2}$, and set $W = UV$. Then $\mathbb{P}(W = \pm 1) = \frac{1}{2}$ and the triple satisfies:

- (i) U and V independent; (ii) U and W independent; (iii) V and W independent.

For (ii), for instance,

$$\mathbb{P}(U = 1, W = 1) = \mathbb{P}(U = 1, V = 1) = \frac{1}{4} = \frac{1}{2} \cdot \frac{1}{2} = \mathbb{P}(U = 1)\mathbb{P}(W = 1),$$

and the other pairs of values are analogous. However the three are not independent, since $W = UV$ imposes a deterministic relation:

$$\mathbb{P}(U = 1, V = 1, W = 1) = \mathbb{P}(U = 1, V = 1) = \frac{1}{4} \neq \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \mathbb{P}(U = 1)\mathbb{P}(V = 1)\mathbb{P}(W = 1).$$

Properties of independent random variables

1. If $X_1, \dots, X_n$ are independent and $g_i : \mathbf{R} \rightarrow \mathbf{R}$ and $\mathbb{E}[|g_i(X_i)|] < \infty$, then

$$\mathbb{E}\left[\prod_{i=1}^n g_i(X_i)\right] = \prod_{i=1}^n \mathbb{E}[g_i(X_i)].$$

In particular, for independent X, Y with expectation, $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$.

2. Define the **covariance** between two random variables as

$$\text{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y],$$

$$\text{Var}(X) = \text{Cov}(X, X).$$

Basic identities (for scalars a, b, c, d):

$$\text{Cov}(aX + b, cY + d) = ac \text{Cov}(X, Y),$$

$$\text{Var}(aX + b) = a^2 \text{Var}(X).$$

3. Let $S_n = \sum_{i=1}^n X_i$ with $\mathbb{E}[X_i^2] < \infty$. Then

$$\text{Var}(S_n) = \sum_{i=1}^n \text{Var}(X_i) + 2 \sum_{1 \leq i < j \leq n} \text{Cov}(X_i, X_j).$$

If the X_i are pairwise uncorrelated ($\text{Cov}(X_i, X_j) = 0$ for $i \neq j$), then

$$\text{Var}(S_n) = \sum_{i=1}^n \text{Var}(X_i).$$

In particular, for independent X_i ,

$$\text{Var}\left(\sum_{i=1}^n X_i\right) = \sum_{i=1}^n \text{Var}(X_i).$$

Example 8 (Uncorrelated but not independent). Let U take values $-1, 0, 1$ with equal probabilities $\frac{1}{3}$. Define

$$X = U, \quad Y = U^2.$$

Then

$$\begin{aligned} \mathbb{E}[X] &= \frac{1}{3}(-1) + \frac{1}{3}(0) + \frac{1}{3}(1) = 0, & \mathbb{E}[Y] &= \frac{1}{3}(1) + \frac{1}{3}(0) + \frac{1}{3}(1) = \frac{2}{3}, \\ \mathbb{E}[XY] &= \mathbb{E}[U \cdot U^2] = \mathbb{E}[U^3] = \frac{1}{3}(-1) + \frac{1}{3}(0) + \frac{1}{3}(1) = 0. \end{aligned}$$

Hence

$$\text{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y] = 0 - 0 \cdot \frac{2}{3} = 0,$$

so X and Y are uncorrelated. They are *not* independent: for instance,

$$\mathbb{P}(Y = 0) = \frac{1}{3}, \quad \mathbb{P}(Y = 0 \mid X = 0) = 1, \quad \mathbb{P}(Y = 0 \mid X = \pm 1) = 0, \quad (\text{找到一个反例即可})$$

so the distribution of Y changes with the value of X .

3 Lecture 4

Standard distributions

Definition 4 (Bernoulli(p)). A random variable B has the Bernoulli(p) distribution if

$$\mathbb{P}(B = 1) = p, \quad \mathbb{P}(B = 0) = 1 - p, \quad p \in [0, 1].$$

Basic facts (quick reminders):

$$\mathbb{E}[B] = 1 \cdot p + 0 \cdot (1 - p) = p, \quad \text{Var}(B) = \mathbb{E}[B^2] - \mathbb{E}[B]^2 = \mathbb{E}[B] - p^2 = p(1 - p),$$

since $B^2 = B$. The variance is maximized at $p = \frac{1}{2}$.

Example 9 (Rademacher from Bernoulli). Let $B \sim \text{Bernoulli}(p)$ and set $R = 2B - 1$. Then $R \in \{-1, 1\}$ with

$$\mathbb{P}(R = 1) = p, \quad \mathbb{P}(R = -1) = 1 - p.$$

Hence $\mathbb{E}[R] = 2p - 1$ and $\text{Var}(R) = 4 \text{Var}(B) = 4p(1 - p)$. The “standard Rademacher” is obtained at $p = \frac{1}{2}$, giving $\mathbb{P}(R = \pm 1) = \frac{1}{2}$, $\mathbb{E}[R] = 0$, $\text{Var}(R) = 1$.

Definition 5 (Binomial(n, p)). A random variable X has the Binomial(n, p) distribution (n is a natural number, $p \in [0, 1]$) if

$$\mathbb{P}(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}, \quad k = 0, 1, \dots, n.$$

One construction: let $B_1, \dots, B_n$ be independent Bernoulli(p) variables, and define

$$X = \sum_{i=1}^n B_i.$$

Then X counts the number of successes in n independent trials, hence has the Binomial(n, p) pmf above (choose which k trials succeed and multiply by $p^k(1-p)^{n-k}$). Expectations via indicators and independence:

$$\mathbb{E}[X] = \sum_{i=1}^n \mathbb{E}[B_i] = np,$$

$$\text{Var}(X) = \sum_{i=1}^n \text{Var}(B_i) = np(1-p).$$

Problem 6 (Urn with replacement, N balls). Consider an urn with N balls. Exactly R of the N balls are red and the remaining $W = N - R$ are white (balls are identical in every other respect). We perform n draws with replacement. What is the probability that exactly w of the n draws are white?

Example 10. For $i = 1, \dots, n$ let $B_i = \mathbf{1}(\text{the } i\text{-th draw is white})$. Then

$$\mathbb{P}(B_i = 1) = \frac{W}{N} = p, \quad \mathbb{P}(B_i = 0) = 1 - p = \frac{R}{N},$$

and replacement makes $B_1, \dots, B_n$ independent. Let $X = \sum_{i=1}^n B_i$ be the number of white draws. Then $X \sim \text{Bin}(n, p)$ and

$$\mathbb{P}(X = w) = \binom{n}{w} \left(\frac{W}{N}\right)^w \left(\frac{R}{N}\right)^{n-w}.$$

Hypergeometric distribution

Definition 6 (Hypergeometric distribution). Let X be the number of white balls in a sample of size n drawn without replacement from an urn with W white and R red balls ($W + R = N$). Then X has the Hypergeom(N, W, n) distribution with support

$$w \in \{\max(0, n - R), \dots, \min(n, W)\}$$

and probability mass function

$$\mathbb{P}(X = w) = \frac{\binom{W}{w} \binom{R}{n-w}}{\binom{N}{n}}.$$

N : 球总数
 W : 目标球总数
 n : 抽取次数

目标球潜在情况数 非目标球潜在情况数 所有潜在情况数

Example 11 (Counting proof of the pmf). All $\binom{N}{n}$ size- n subsets are equally likely. To obtain exactly w white and $n - w$ red draws, choose w from the W white and $n - w$ from the R red:

$$\binom{W}{w} \binom{R}{n-w}.$$

Divide by $\binom{N}{n}$ to get $\mathbb{P}(X = w) = \frac{\binom{W}{w} \binom{R}{n-w}}{\binom{N}{n}}.$



Hypergeometric setup: $X \sim \text{Hypergeom}(N, W, n)$ counts white balls in n draws without replacement from an urn with W white and $R = N - W$ red. Let $p := W/N$. Introduce indicators $B_i := \mathbf{1}(i\text{-th draw is white})$ for $i = 1, \dots, n$, and write

$$X = \sum_{i=1}^n B_i.$$

Mean: By symmetry,

$$\mathbb{E}[B_i] = \mathbb{P}(B_i = 1) = \frac{W}{N} = p, \quad \text{therefore} \quad \mathbb{E}[X] = \sum_{i=1}^n \mathbb{E}[B_i] = np.$$

Variance: First,

$$\text{Var}(B_i) = p(1-p).$$

For $i \neq j$,

$$\mathbb{E}[B_i B_j] = \mathbb{P}(B_i = 1, B_j = 1) = \frac{W}{N} \cdot \frac{W-1}{N-1} \quad \text{therefore} \quad \text{Cov}(B_i, B_j) = \mathbb{E}[B_i B_j] - p^2 = -\frac{p(1-p)}{N-1}.$$

Therefore,

$$\text{Var}(X) = \sum_{i=1}^n \text{Var}(B_i) + 2 \sum_{1 \leq i < j \leq n} \text{Cov}(B_i, B_j) = np(1-p) + 2 \binom{n}{2} \left(-\frac{p(1-p)}{N-1} \right) = np(1-p) \frac{N-n}{N-1}.$$

3.1 Binomial approximation of the hypergeometric distribution

Approximation of the hypergeometric by a binomial

Let $N = R + W$ (reds R , whites W) and draw $n = r + w$ balls without replacement. The hypergeometric probability of seeing exactly r reds and w whites is

$$\frac{\binom{R}{r} \binom{W}{w}}{\binom{N}{n}} = \frac{\binom{R}{r} \binom{W}{w}}{\binom{N}{r+w}} = \binom{n}{r} \frac{R(R-1) \cdots (R-r+1)}{N(N-1) \cdots (N-r+1)} \frac{W(W-1) \cdots (W-w+1)}{(N-r)(N-r-1) \cdots (N-n+1)}.$$

Equivalently, using products,

$$\frac{\binom{R}{r} \binom{W}{w}}{\binom{N}{n}} = \binom{n}{r} \prod_{i=0}^{r-1} \frac{R-i}{N-i} \prod_{j=0}^{w-1} \frac{W-j}{N-r-j}.$$

Assume N is much larger than n (so i/N and $(r+j)/N$ are small). Then

$$\frac{R-i}{N-i} = \frac{R/N - i/N}{1 - i/N} \approx \frac{R}{N}, \quad \frac{W-j}{N-r-j} = \frac{W/N - j/N}{1 - (r+j)/N} \approx \frac{W}{N}.$$

Hence

$$\frac{\binom{R}{r} \binom{W}{w}}{\binom{N}{n}} \approx \binom{n}{r} \left(\frac{R}{N}\right)^r \left(\frac{W}{N}\right)^w.$$

With $p := R/N$ (so $1 - p = W/N$) and $n = r + w$:

$$\mathbb{P}(X = r) \approx \binom{n}{r} p^r (1 - p)^{n-r},$$

i.e., the hypergeometric $\text{Hypergeom}(N, W, n)$ is well-approximated by $\text{Bin}(n, p)$ when $n \ll N$.

• Geometric
• Poisson
• PGF

4 Lecture 5

Geometric distribution

A random variable X has the Geometric(p) distribution ($p \in [0, 1]$) if it takes values $1, 2, \dots$ with

$$\mathbb{P}(X = i) = (1 - p)^{i-1} p, \quad i = 1, 2, \dots$$

Interpretation: X is the number of independent Bernoulli(p) trials needed to obtain the first success.

Example 12 (Deriving the probability mass function from Bernoulli trials). Let $B_1, B_2, \dots$ be i.i.d. Bernoulli(p), where $B_t = 1$ means success on trial t . The event $\{X = i\}$ is $\{B_1 = 0, \dots, B_{i-1} = 0, B_i = 1\}$, whose probability is $(1 - p)^{i-1} p$ by independence.

Let's compute mean and variance by series calculus. Write $q = 1 - p$. Then

$$\mathbb{P}(X = i) = q^{i-1} p.$$

Use the standard power-series identities (for $|q| < 1$):

$$\sum_{i=1}^{\infty} i q^{i-1} = \frac{1}{(1 - q)^2}, \quad \sum_{i=1}^{\infty} i^2 q^{i-1} = \frac{1 + q}{(1 - q)^3}.$$

Therefore

$$\mathbb{E}[X] = \sum_{i=1}^{\infty} i \mathbb{P}(X = i) = p \sum_{i=1}^{\infty} i q^{i-1} = p \cdot \frac{1}{(1 - q)^2} = \frac{1}{p},$$

$$\mathbb{E}[X^2] = \sum_{i=1}^{\infty} i^2 \mathbb{P}(X = i) = p \sum_{i=1}^{\infty} i^2 q^{i-1} = p \cdot \frac{1 + q}{(1 - q)^3} = \frac{2 - p}{p^2}.$$

Hence

$$\text{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2 = \frac{2 - p}{p^2} - \frac{1}{p^2} = \frac{1 - p}{p^2}.$$

$$\begin{aligned} \mathbb{E}(X) &= \sum_{x=1}^{\infty} x \underbrace{(1-p)^{x-1} p}_{p(x)} \\ &= p \sum_{x=1}^{\infty} x q^{x-1} \quad (\text{let } q = 1 - p) \\ &= p \sum_{x=1}^{\infty} \frac{d}{dq} q^x \\ &= p \frac{d}{dq} \sum_{x=1}^{\infty} q^x \\ &= p \frac{d}{dq} \left(\frac{q}{1 - q} \right) \\ &= p \frac{d}{dq} \left(\frac{1}{1 - q} - 1 \right) \\ &= \frac{p}{(1 - q)^2} = \frac{p}{p^2} = \frac{1}{p} \end{aligned}$$

$$\begin{aligned} &\sum_{i=1}^{\infty} (i+1) i q^{i-1} \\ &= \frac{\partial}{\partial q^2} \sum_{i=1}^{\infty} q^{i+1} \\ &= \frac{\partial}{\partial q^2} \frac{q^2}{1 - q} \\ &= \frac{2}{(1 - q)^3} \\ &\Rightarrow \sum_{i=1}^{\infty} i^2 q^{i-1} = \frac{2}{(1 - q)^3} - \frac{1}{(1 - q)^2} \\ &= \frac{1 + q}{(1 - q)^3} \end{aligned}$$

The memorylessness property of geometric distribution. Tail probabilities: $\mathbb{P}(X > n) = q^n$. Then for all integers $m, n \geq 0$,

$$\mathbb{P}(X > m + n \mid X > n) = \frac{\mathbb{P}(X > m + n)}{\mathbb{P}(X > n)} = \frac{q^{m+n}}{q^n} = q^m = \mathbb{P}(X > m),$$

so $\text{Geometric}(p)$ is “memoryless”.

Poisson distribution

Definition 7 ($\text{Poisson}(\lambda)$). A discrete random variable X has the $\text{Poisson}(\lambda)$ distribution with intensity $\lambda > 0$ if

$$\mathbb{P}(X = k) = e^{-\lambda} \frac{\lambda^k}{k!}, \quad k = 0, 1, 2, \dots$$

Mean:

$$\mathbb{E}[X] = \sum_{k \geq 0} k e^{-\lambda} \frac{\lambda^k}{k!} = e^{-\lambda} \lambda \sum_{k \geq 1} \frac{\lambda^{k-1}}{(k-1)!} = \lambda.$$

Consider

$$\mathbb{E}[X(X-1)] = \sum_{k \geq 0} k(k-1) e^{-\lambda} \frac{\lambda^k}{k!} = e^{-\lambda} \lambda^2 \sum_{k \geq 2} \frac{\lambda^{k-2}}{(k-2)!} = \lambda^2.$$

为了构造 pgf

Thus, we have

$$\text{Var}(X) = \mathbb{E}[X(X-1)] + \mathbb{E}[X] - \mathbb{E}[X]^2 = \lambda^2 + \lambda - \lambda^2 = \lambda.$$

Sum stability: if $X_1, \dots, X_m$ are independent with $X_i \sim \text{Poi}(\lambda_i)$, then

$$\sum_{i=1}^m X_i \sim \text{Poi}\left(\sum_{i=1}^m \lambda_i\right),$$

Probability generating function (PGF) and its properties. For a nonnegative integer-valued r.v. X define

$$G_X(s) := \mathbb{E}[s^X], \quad s \in [0, 1].$$

Basic facts (all follow from $G_X(s) = \sum_{k \geq 0} \mathbb{P}(X = k) s^k$):

- $G_X(1) = 1$, $G_X(0) = \mathbb{P}(X = 0)$.

- Coefficients are recovered by derivatives at 0:

$$\mathbb{P}(X = k) = \frac{G_X^{(k)}(0)}{k!}, \quad k = 0, 1, 2, \dots$$

so the pgf determines the distribution of X .

- Moments from derivatives at 1 (when finite):

$$\mathbb{E}[X] = G'_X(1), \quad \text{Var}(X) = G''_X(1) + G'_X(1) - (G'_X(1))^2.$$

- If X, Y are independent, then

$$G_{X+Y}(s) = \mathbb{E}[s^{X+Y}] = \mathbb{E}[s^X s^Y] = \mathbb{E}[s^X] \mathbb{E}[s^Y] = G_X(s) G_Y(s).$$

(The factorization uses independence; the restriction $s \in [0, 1]$ ensures all expectations exist.)

• PGF的应用

5 Lecture 6

Poisson PGF and sum stability. If $X \sim \text{Poi}(\lambda)$, then

$$G_X(s) = \sum_{k \geq 0} e^{-\lambda} \frac{\lambda^k}{k!} s^k = e^{-\lambda} \exp(\lambda s) = \exp(\lambda(s - 1)), \quad s \in [0, 1].$$

Let $X_1, \dots, X_m$ be independent with $X_i \sim \text{Poi}(\lambda_i)$. Using the product rule,

$$G_{\sum_{i=1}^m X_i}(s) = \prod_{i=1}^m G_{X_i}(s) = \prod_{i=1}^m \exp(\lambda_i(s - 1)) = \exp\left(\left(\sum_{i=1}^m \lambda_i\right)(s - 1)\right).$$

The right-hand side is the pgf of $\text{Poi}(\sum_{i=1}^m \lambda_i)$, thus

$$\sum_{i=1}^m X_i \stackrel{d}{=} \text{Poi}\left(\sum_{i=1}^m \lambda_i\right).$$



Problem 7 (Two independent streams: TTH vs HHT (PGF)). Each round Alice and Bob flip their *own* fair coins independently. Let T_A be Alice's waiting time to the pattern TTH , and T_B Bob's waiting time to HHT .

- Use probability generating functions to show that T_A and T_B have the same distribution; conclude that with a fair tie-break the game is fair.
- Using the same PGF, compute the *unconditional* expected number of rounds until Alice sees her winning pattern TTH .

Example 13 (PGF solution). While revealing the coin sequence, we do not need to remember the whole past. To know how close we are to the target pattern TTH , it suffices to keep the *longest suffix of the revealed string that is also a prefix of TTH* . This produces three transient “progress” states and one absorbing success:

$\emptyset$ (no match yet), T (matched the first T), TT (matched TT), $\boxed{TTH}$ (absorb).

State updates:

- From $\emptyset$: seeing T starts a match $\emptyset \rightarrow T$; seeing H gives no prefix, so $\emptyset \rightarrow \emptyset$.
- From T : if the next toss is T , we extend to TT ; if it is H , the longest suffix that is again a prefix of TTH is just $\emptyset$ (no match), so $T \rightarrow \emptyset$.
- From TT : if the next toss is H we complete TTH (absorb); if it is T , the longest suffix that is a prefix is still TT (the last two T 's), so $TT \rightarrow TT$.

Probability generating functions. For a state $s \in \{\emptyset, T, TT\}$, let $G_s(z)$ be the pgf of the time (number of further tosses) to hit TTH starting from s . Each step multiplies by z ; an absorbing hit contributes the constant 1 (we stop). With a fair coin,

$$\begin{aligned} G_{TT}(z) &= \frac{1}{2}z \cdot \underbrace{1}_{\text{toss } T} + \frac{1}{2}z G_{TT}(z) && \text{(from } TT: H \text{ wins, } T \text{ loops),} \\ G_T(z) &= \frac{1}{2}z G_{TT}(z) + \frac{1}{2}z G_{\emptyset}(z) && \text{(from } T: T \rightarrow TT, H \rightarrow \emptyset), \\ G_{\emptyset}(z) &= \frac{1}{2}z G_T(z) + \frac{1}{2}z G_{\emptyset}(z) && \text{(from } \emptyset: T \rightarrow T, H \rightarrow \emptyset). \end{aligned}$$

Solve the system. From the first equation,

$$G_{TT}(z) = \frac{z}{2-z}.$$

Substitute into the second, then use the third to eliminate $G_{\emptyset}$:

$$G_{\emptyset}(z) = \frac{z^3}{z^3 - 8(z-1)}.$$

Symmetry for HHT . If we swap $T \leftrightarrow H$, the same automaton and algebra apply; hence the pgf for the waiting time to HHT is the same:

$$G_{HHT}(z) = \frac{z^3}{z^3 - 8(z-1)} = G_{TTH}(z) =: G(z).$$

Therefore $T_A \stackrel{d}{=} T_B$; with a fair tie-break the two-stream game is fair.

From $G(z) = z^3/(z^3 - 8z + 8)$,

$$G'(1) = \mathbb{E}[T_A] = 8.$$

So, when Alice flips her own fair coin each round, the expected number of rounds until she sees TTH is 8.



Problem 8 (One shared coin: TTH vs THH (exact recursion)). A single fair coin is tossed once per step and both players watch the same sequence. Alice wins if the first length-3 pattern to appear is TTH . Bob wins if the first length-3 pattern to appear is THH . Compute $\mathbb{P}(TTH \text{ appears before } THH)$. Is the game fair?

Example 14 (Suffix-prefix recursion). We only need to remember how much of either target has been matched at the end of the current prefix. Keep the *longest suffix that is also a prefix* of TTH or THH . The relevant (non-absorbing) states are

$$\emptyset, \quad T, \quad TT, \quad TH,$$

and the two absorbing states are TTH (Alice wins) and THH (Bob wins).

State updates (fair coin).

- From $\emptyset$: $T \rightarrow T$, $H \rightarrow \emptyset$.
- From T : $T \rightarrow TT$, $H \rightarrow TH$.
- From TT : $H \rightarrow \text{absorb at } TTH$; $T \rightarrow TT$.
- From TH : $H \rightarrow \text{absorb at } THH$; $T \rightarrow T$.

Let

$$p_s = \mathbb{P}(TTH \text{ occurs before } THH \mid \text{current suffix } s).$$

Using the transitions above and that each next toss has probability $1/2$, we obtain

$$\begin{aligned} p_{TT} &= \frac{1}{2} \cdot 1 + \frac{1}{2} p_{TT}, \\ p_{TH} &= \frac{1}{2} p_T + \frac{1}{2} \cdot 0, \\ p_T &= \frac{1}{2} p_{TT} + \frac{1}{2} p_{TH}, \\ p_{\emptyset} &= \frac{1}{2} p_T + \frac{1}{2} p_{\emptyset}. \end{aligned}$$

From the first equation, $p_{TT} = 1$. Then $p_{TH} = \frac{1}{2} p_T$, and

$$p_T = \frac{1}{2} \cdot 1 + \frac{1}{2} \cdot \frac{1}{2} p_T \implies p_T - \frac{1}{4} p_T = \frac{1}{2} \implies p_T = \frac{2}{3}.$$

Finally,

$$p_{\emptyset} = \frac{1}{2} p_T + \frac{1}{2} p_{\emptyset} \implies p_{\emptyset} = p_T = \frac{2}{3}.$$

Finally,

$$\mathbb{P}(TTH \text{ appears before } THH) = \boxed{\frac{2}{3}}, \quad \mathbb{P}(THH \text{ appears before } TTH) = \boxed{\frac{1}{3}}.$$

Thus, for this shared-coin match-up the game is *not* fair: Alice's pattern TTH wins with probability $2/3$.

• CDF
 • Continuous r.v.
 • Uniform
 • Normal
 • Exponential
 • Bin(n,p) is normal limit
 (De Moivre-Laplace CLT)

6 Lecture 7

Cumulative distribution function (CDF)

For any real-valued random variable X (discrete or continuous), the *cumulative distribution function* is

$$F(x) := \mathbb{P}(X \leq x), \quad -\infty < x < \infty.$$

Basic properties of every cdf:

- F is nondecreasing and right-continuous;
- $\lim_{x \rightarrow -\infty} F(x) = 0$ and $\lim_{x \rightarrow \infty} F(x) = 1$.

If X admits a density f , then

$$F(x) = \int_{-\infty}^x f(t) dt, \quad \text{and} \quad f(x) = F'(x) \text{ for a.e. } x.$$

Continuous random variables and densities

A (real-valued) random variable X is *continuous* if there exists a function $f : \mathbf{R} \rightarrow [0, \infty)$ (the *probability density function*, pdf) such that for every Borel set $A \subset \mathbf{R}$

$$\mathbb{P}(X \in A) = \int_A f(x) dx, \quad \text{with} \quad \int_{-\infty}^{\infty} f(x) dx = 1.$$

In particular, for any $a < b$,

$$\mathbb{P}(a \leq X \leq b) = \int_a^b f(x) dx, \quad \mathbb{P}(X = x) = \int_x^x f(t) dt = 0.$$

A useful local interpretation (when f is continuous at x): for small $\delta > 0$,

$$\mathbb{P}(x \leq X \leq x + \delta) \approx f(x) \delta.$$

For a function $g : \mathbf{R} \rightarrow \mathbf{R}$ with $\int |g(x)| f(x) dx < \infty$,

$$\mathbb{E}[g(X)] = \int_{-\infty}^{\infty} g(x) f(x) dx.$$

In particular, $\mathbb{E}[X] = \int x f(x) dx$ and

$$\text{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2] = \int (x - \mathbb{E}[X])^2 f(x) dx = \mathbb{E}[X^2] - \mathbb{E}[X]^2, \quad \mathbb{E}[X^2] = \int x^2 f(x) dx.$$

Uniform distribution

Definition 8 (Uniform(a, b)). For $a < b$, $U \sim U(a, b)$ has density

$$f_U(x) = \begin{cases} \frac{1}{b-a}, & a < x < b, \\ 0, & \text{otherwise.} \end{cases}$$

Using the formulas above,

$$\mathbb{E}[U] = \int_a^b \frac{x}{b-a} dx = \frac{a+b}{2}, \quad \mathbb{E}[U^2] = \int_a^b \frac{x^2}{b-a} dx = \frac{a^2 + ab + b^2}{3},$$

hence

$$\text{Var}(U) = \mathbb{E}[U^2] - \mathbb{E}[U]^2 = \frac{(b-a)^2}{12}.$$

Normal (Gaussian) distribution

Definition 9 (Normal $N(\mu, \sigma^2)$). A random variable X has the normal distribution with mean $\mu \in \mathbf{R}$ and variance $\sigma^2 > 0$ if

$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \quad x \in \mathbf{R}.$$

The case $\mu = 0$, $\sigma^2 = 1$ is the standard normal, with density $\phi(x) = \frac{1}{\sqrt{2\pi}}e^{-x^2/2}$.

Basic facts:

$$\mathbb{E}[X] = \mu, \quad \text{Var}(X) = \sigma^2.$$

Standardization: if $Z = \frac{X - \mu}{\sigma}$, then $Z \sim N(0, 1)$; conversely $X = \mu + \sigma Z$.

Exponential distribution

Definition 10 (Exponential(λ)). For a rate $\lambda > 0$, $X \sim \text{Exp}(\lambda)$ has density

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0, \\ 0, & x < 0, \end{cases} \quad \mathbb{P}(X > x) = e^{-\lambda x}, \quad x \geq 0.$$

Using the continuous formulas,

$$\mathbb{E}[X] = \int_0^\infty x \lambda e^{-\lambda x} dx = \frac{1}{\lambda}, \quad \mathbb{E}[X^2] = \int_0^\infty x^2 \lambda e^{-\lambda x} dx = \frac{2}{\lambda^2},$$

so

$$\text{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2 = \frac{1}{\lambda^2}.$$

Entropy approximation of $\text{Bin}(n, p)$ and the normal limit (second-order)

Let $X \sim \text{Bin}(n, p)$ and write $f := k/n \in (0, 1)$ for the empirical fraction of successes. Using Stirling's asymptotic $m! \sim (m/e)^m \sqrt{2\pi m}$ in

$$\mathbb{P}(X = k) = \binom{n}{k} p^k (1-p)^{n-k},$$

$\binom{n}{k} = \frac{n!}{k!(n-k)!}$

we obtain

$$\begin{aligned} \mathbb{P}(X = k) &\approx \frac{1}{\sqrt{2\pi}} \sqrt{\frac{n}{k(n-k)}} \frac{n^n}{k^k (n-k)^{n-k}} p^k (1-p)^{n-k} \\ &= \frac{1}{\sqrt{2\pi n f(1-f)}} \exp\left\{ -n \left[f \log \frac{f}{p} + (1-f) \log \frac{1-f}{1-p} \right] \right\}. \end{aligned}$$

Introduce the *relative entropy* (binary Kullback-Leibler divergence)

$$D(f\|p) = f \log \frac{f}{p} + (1-f) \log \frac{1-f}{1-p} \geq 0,$$

so the Stirling/entropy approximation reads

$$\mathbb{P}(X = k) \approx \frac{1}{\sqrt{2\pi n f(1-f)}} \exp(-n D(f\|p)).$$

Second-order Taylor expansion around $f = p$. Set $G(f) := D(f\|p)$. Then

$$G'(f) = \log \frac{f}{p} - \log \frac{1-f}{1-p}, \quad G''(f) = \frac{1}{f} + \frac{1}{1-f}.$$

In particular, $G(p) = 0$, $G'(p) = 0$, and $G''(p) = \frac{1}{p(1-p)}$. Thus for f close to p ,

$$G(f) \approx \frac{G''(p)}{2} (f-p)^2 = \frac{(f-p)^2}{2p(1-p)}.$$

Plugging this into the entropy approximation and replacing f by p in the pre-factor (valid when f is near p), we obtain the **de Moivre–Laplace normal approximation**

$$\mathbb{P}(X = k) \approx \frac{1}{\sqrt{2\pi np(1-p)}} \exp\left(-\frac{(k-np)^2}{2np(1-p)}\right).$$

Fact (De Moivre–Laplace Central Limit Theorem). Let $X \sim \text{Bin}(n, p)$ with fixed $0 < p < 1$. For every fixed pair $a < b$,

$$\lim_{n \rightarrow \infty} \mathbb{P}\left\{ a \leq \frac{X - np}{\sqrt{np(1-p)}} \leq b \right\} = \int_a^b \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{z^2}{2}\right) dz.$$

- KL 散度
- Markov ineq.
- Chebyshev ineq.

7 Lecture 8

Relative entropy (Kullback–Leibler divergence)

We write $D(P\|Q)$ for the *Kullback–Leibler (KL) divergence* between two probability distributions P and Q .

Definition 11 (Discrete KL). Let P, Q be distributions on a countable set $\mathcal{X}$ with pmfs p, q . If there exists x with $p(x) > 0$ and $q(x) = 0$, set $D(P\|Q) = +\infty$. Otherwise,

$$D(P\|Q) = \sum_{x \in \mathcal{X}} p(x) \log \frac{p(x)}{q(x)}.$$

Definition 12 (Continuous KL). Let f, g be two densities. Set

$$D(f\|g) = \int_{\mathbf{R}} f(x) \log \frac{f(x)}{g(x)} dx.$$

Importantly, $D(f\|g) = +\infty$ when $g(x) = 0$ on some set where $f(x) > 0$.

Example 15 (Binary/Bernoulli case). For $P = \text{Bernoulli}(f)$ and $Q = \text{Bernoulli}(p)$,

$$D(\text{Be}(f)\|\text{Be}(p)) = f \log \frac{f}{p} + (1-f) \log \frac{1-f}{1-p}.$$

This is the “binary divergence” used earlier with $f = k/n$. It is usually denoted just $D(f\|p)$.

Example 16 (Poisson KL: quick derivation). Let $P = \text{Poi}(\lambda)$ and $Q = \text{Poi}(\lambda_0)$ on $\{0, 1, 2, \dots\}$ with

$$p(k) = e^{-\lambda} \frac{\lambda^k}{k!}, \quad q(k) = e^{-\lambda_0} \frac{\lambda_0^k}{k!}.$$

Then, we have

$$\log \frac{p(k)}{q(k)} = (-\lambda + k \log \lambda - \log k!) - (-\lambda_0 + k \log \lambda_0 - \log k!) = \lambda_0 - \lambda + k \log \frac{\lambda}{\lambda_0}.$$

Finally, ~~~~~ 先处理这一部分

$$\begin{aligned} D(P\|Q) &= \sum_{k=0}^{\infty} p(k) \log \frac{p(k)}{q(k)} = \sum_{k=0}^{\infty} p(k) \left(\lambda_0 - \lambda + k \log \frac{\lambda}{\lambda_0} \right) \\ &= (\lambda_0 - \lambda) \sum_k p(k) + \log \frac{\lambda}{\lambda_0} \sum_k k p(k) = \lambda_0 - \lambda + \log \frac{\lambda}{\lambda_0} \mathbb{E}_P[K]. \end{aligned}$$

~~~~~ = 1~~~~~ = E\_P(K)

For  $K \sim \text{Poi}(\lambda)$ ,  $\mathbb{E}_P[K] = \lambda$ , so we can write

$$D(\text{Poi}(\lambda)\|\text{Poi}(\lambda_0)) = \lambda \log \frac{\lambda}{\lambda_0} + \lambda_0 - \lambda.$$

*Jensen's inequality (for random variables).* If  $\varphi : \mathbf{R} \rightarrow \mathbf{R}$  is convex and  $Y$  is integrable, then

$$\mathbb{E}[\varphi(Y)] \geq \varphi(\mathbb{E}[Y]).$$

Equality holds if and only if  $Y$  is a constant or  $\varphi$  is linear on the range of  $Y$ .

**Divergence is non-negative.** We start with the discrete case.

$$D(P\|Q) = \sum_x p(x) \log \frac{p(x)}{q(x)} = \mathbb{E}_P \left[ \log \frac{p(X)}{q(X)} \right] = \mathbb{E}_P \left[ -\log \frac{q(X)}{p(X)} \right].$$

Since  $-\log$  is convex, Jensen gives

$$D(P\|Q) \geq -\log \mathbb{E}_P \left[ \frac{q(X)}{p(X)} \right] = -\log \sum_x q(x) = -\log 1 = 0.$$

Moreover  $D(P\|Q) = 0$  iff  $p(x) = q(x)$  for all  $x$  with  $p(x) > 0$  (i.e.,  $P = Q$ ).

Similarly, in the continuous case

$$D(f\|g) = \int f(x) \log \frac{f(x)}{g(x)} dx = \mathbb{E}_f \left[ -\log \frac{g(X)}{f(X)} \right] \geq -\log \mathbb{E}_f \left[ \frac{g(X)}{f(X)} \right] = -\log \int g(x) dx = 0.$$

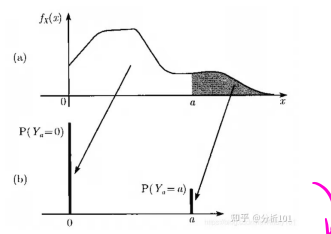
*Basic properties of divergence:*

- **Nonnegativity:**  $D(P\|Q) \geq 0$ ; equality iff  $P = Q$  (a.e.).
- **Asymmetry:**  $D(P\|Q) \neq D(Q\|P)$  in general.
- Not a metric (violates symmetry and triangle inequality).

## Basic tail inequalities

**Markov's inequality.** If  $X \geq 0$  and  $t > 0$ , then

$$\mathbb{P}(X \geq t) \leq \frac{\mathbb{E}[X]}{t}.$$



*Proof.* For a nonnegative  $X$  and  $t > 0$  we have the pointwise bound

$$X \geq t \mathbf{1}(X \geq t).$$

Taking expectations gives

$$\mathbb{E}[X] \geq t \mathbb{E}[\mathbf{1}(X \geq t)] = t \mathbb{P}(X \geq t).$$

Dividing by  $t$  yields the desired bound  $\mathbb{P}(X \geq t) \leq \mathbb{E}[X]/t$ .

**Chebyshev's inequality.** If  $X$  has finite variance and  $t > 0$ , then

$$\mathbb{P}(|X - \mathbb{E}[X]| \geq t) \leq \frac{\text{Var}(X)}{t^2}.$$

*Proof.* We apply Markov's inequality to  $(X - \mathbb{E}[X])^2$ . We get  $\mathbb{P}((X - \mathbb{E}[X])^2 \geq t^2) \leq \mathbb{E}[(X - \mathbb{E}[X])^2]/t^2$ .

- Chebyshev LLN
- Gaussian tail
- MGF
- Chernoff's method

## 8 Lecture 9

**Fact. Chebyshev's law of large numbers (quantitative).** Let  $X_1, \dots, X_n$  be independent with the same mean  $\mu$  and variance  $\sigma^2 < \infty$ . Define the sample average

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i.$$

Then for every  $t > 0$ ,

$$\mathbb{P}(|\bar{X}_n - \mu| \geq t) \leq \frac{\sigma^2}{n t^2}.$$

In words: as  $n$  grows, the chance that the average differs from  $\mu$  by more than  $t$  is at most  $\sigma^2/(n t^2)$ , which goes to 0 at rate  $1/n$ .

*Proof.* By linearity of expectation it holds that  $\mathbb{E}[\bar{X}_n] = \mu$ . Using independence, we have

$$\text{Var}(\bar{X}_n) = \text{Var}\left(\frac{1}{n} \sum_{i=1}^n X_i\right) = \frac{1}{n^2} \sum_{i=1}^n \text{Var}(X_i) = \frac{\sigma^2}{n}.$$

Applying Chebyshev's inequality to  $\bar{X}_n$  we get

$$\mathbb{P}(|\bar{X}_n - \mu| \geq t) \leq \frac{\text{Var}(\bar{X}_n)}{t^2} = \frac{\sigma^2}{n t^2}.$$

**Gaussian tails.** Let  $Z \sim N(0, 1)$  and  $\phi(t) = \frac{1}{\sqrt{2\pi}} e^{-t^2/2}$ . For any  $t \geq 0$ ,

$$\left( \frac{1}{t} - \frac{1}{t^3} \right) \phi(t) \leq \mathbb{P}(Z \geq t) \leq \frac{1}{t} \phi(t).$$

*Proof.* Write

$$\mathbb{P}(Z \geq t) = \frac{1}{\sqrt{2\pi}} \int_t^\infty e^{-x^2/2} dx.$$

For the upper bound we set  $x = t + y$ ,

$$\int_t^\infty e^{-x^2/2} dx = \int_0^\infty e^{-t^2/2 - \underbrace{ty - y^2/2}_{<0}} dy \leq e^{-t^2/2} \underbrace{\int_0^\infty e^{-ty} dy}_{-\frac{1}{t} e^{-ty} \Big|_0^\infty = \frac{1}{t}} = \frac{e^{-t^2/2}}{t}.$$

For the lower bound we observe

$$\frac{d}{dx} \left( e^{-x^2/2} \left( \frac{1}{x} - \frac{1}{x^3} \right) \right) = -(1 - 3x^{-4})e^{-x^2/2}.$$

Integrating from  $t$  to  $\infty$  gives

$$\int_t^\infty (1 - 3x^{-4})e^{-x^2/2} dx = \left( \frac{1}{t} - \frac{1}{t^3} \right) e^{-t^2/2}.$$

For  $t > 0$  we have  $\int_t^\infty e^{-x^2/2} dx \geq \int_t^\infty (1 - 3x^{-4})e^{-x^2/2} dx$ , which yields the lower bound.

## Moment generating functions (MGFs)

**Definition 13** (MGF). For a real r.v.  $X$ , the *moment generating function* (mgf) is

$$M_X(\lambda) = \mathbb{E}[e^{\lambda X}], \quad \lambda \in \mathbf{R},$$

whenever the expectation is finite (usually in a neighborhood of  $\lambda = 0$ ).

*Basic facts.*

- If  $M_X$  is finite in a neighborhood of 0, then  $M_X$  is infinitely differentiable there and

$$M_X^{(k)}(0) = \mathbb{E}[X^k], \quad M_X(\lambda) = \sum_{k=0}^{\infty} \frac{\mathbb{E}[X^k]}{k!} \lambda^k \quad (\text{Taylor at 0}).$$

- If  $X$  and  $Y$  are independent, then  $M_{X+Y}(\lambda) = M_X(\lambda)M_Y(\lambda)$ .
- If  $M_X$  exists in a neighborhood of 0 (say, some finite interval), it uniquely determines the distribution of  $X$  (uniqueness).

**Example 17** (MGF does not have to exist in the neighborhood of zero). Let  $X = \exp(Z)$ , where  $Z \sim \mathcal{N}(0, 1)$ . Observe that for  $\lambda > 0$ ,

$$\mathbb{E}[e^{\lambda e^Z}] = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \exp(\lambda \exp(x) - x^2/2\sigma^2) dx \geq \frac{1}{\sqrt{2\pi}} \int_0^{+\infty} \exp(\lambda \exp(x) - x^2/2\sigma^2) dx = +\infty.$$

$\rightarrow \infty$  as  $x \rightarrow \infty$  for  $\lambda > 0$

MGF的一个应用: Chernoff's method (for  $N(0, \sigma^2)$ )

令  $X \sim N(0, \sigma^2)$ ,  $t > 0$ ,  $\lambda > 0$

则  $X$  的 MGF 为  $E_X(\exp(\lambda X)) = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}\sigma} \exp(\lambda x - \frac{x^2}{2\sigma^2}) dx = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}\sigma} \exp(-\frac{(x-\lambda\sigma^2)^2}{2\sigma^2}) dx \cdot \exp(\frac{\lambda^2\sigma^2}{2}) = \exp(\frac{\lambda^2\sigma^2}{2})$

则 Chernoff's method for  $N(0, \sigma^2)$  为

$$P(X \geq t) = P(\lambda X \geq \lambda t)$$

$$= P(\exp(\lambda X) \geq \exp(\lambda t)) \quad (\text{确保 } \exp(\lambda X) > 0, \text{ 可使用 Markov inequality})$$

$$\leq \frac{E_X[\exp(\lambda X)]}{\exp(\lambda t)} \quad (\text{Chernoff's method})$$

$$= \exp(\frac{\lambda^2\sigma^2}{2} - \lambda t) \quad (\text{代入 } X \text{ 的 MGF})$$

我们希望 bound 越 tight 越好, 因此 minimize RHS: choose  $\lambda^* = \frac{t}{\sigma^2}$

$$P(X \geq t) \leq \exp(-\frac{t^2}{2\sigma^2})$$

注: 若  $\sigma=1$ , 可以比较

$$\cdot \text{Chernoff's method: } P(X \geq t) \leq \exp(-\frac{t^2}{2\sigma^2})$$

$$\cdot \text{Chebyshev's inequality: } P(|X| \geq t) \leq \frac{1}{t^2}$$

$$\cdot \text{Gaussian tail: } P(X \geq t) \leq \frac{1}{\sqrt{2\pi}t} \exp(-\frac{t^2}{2}) \quad (\text{收敛最快, 但形式最复杂})$$

## 9 Lecture 10

**Example 18** (Gaussian mgf and why it is quadratic in  $\lambda$ ). Let  $X \sim N(0, \sigma^2)$ . Then, for any  $\lambda \in \mathbb{R}$ ,

$$\begin{aligned} M_X(\lambda) &= \mathbb{E}[e^{\lambda X}] = \frac{1}{\sqrt{2\pi}\sigma} \int_{\mathbb{R}} \exp\left(\lambda x - \frac{x^2}{2\sigma^2}\right) dx \\ &= \frac{1}{\sqrt{2\pi}\sigma} \int_{\mathbb{R}} \exp\left(-\frac{(x - \sigma^2\lambda)^2}{2\sigma^2}\right) \exp\left(\frac{\sigma^2\lambda^2}{2}\right) dx = \exp\left(\frac{\sigma^2\lambda^2}{2}\right). \end{aligned}$$

The last step uses “completing the square” and that the integral of a centered  $N(0, \sigma^2)$  density equals 1. Hence the mgf has the *quadratic* exponent

$$M_X(\lambda) = \exp\left(\frac{\sigma^2\lambda^2}{2}\right), \quad \log M_X(\lambda) = \frac{\sigma^2}{2}\lambda^2.$$

Intuitively, for a centered Gaussian all odd moments vanish, so the Taylor series of  $M_X$  contains only even powers and sums to a pure quadratic in the exponent.

**Moments and Taylor expansion.** From  $M_X(\lambda) = \exp(\sigma^2\lambda^2/2)$  we have

$$M'_X(0) = \mathbb{E}[X] = 0, \quad M''_X(0) = \mathbb{E}[X^2] = \sigma^2.$$

### Hoeffding's inequality for bounded variables

**Hoeffding's bound.** Let  $X_1, \dots, X_n$  be independent with  $a \leq X_i \leq b$  (with probability 1), and write

$$\mu_i = \mathbb{E}[X_i], \quad \bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i, \quad \mu = \frac{1}{n} \sum_{i=1}^n \mu_i.$$

Then for all  $t > 0$ ,

$$\mathbb{P}(\bar{X}_n - \mu \geq t) \leq \exp\left(-\frac{2nt^2}{(b-a)^2}\right), \quad \mathbb{P}(|\bar{X}_n - \mu| \geq t) \leq 2 \exp\left(-\frac{2nt^2}{(b-a)^2}\right).$$

The classical sharp Hoeffding bound has exponent  $-\frac{2nt^2}{(b-a)^2}$ ; the proof below yields a slightly weaker constant.

*Proof* Let  $S = \sum_{i=1}^n (X_i - \mu_i) = n(\bar{X}_n - \mu)$ . For  $\lambda > 0$ , by Markov's inequality,

$$\mathbb{P}(S \geq ns) \leq e^{-\lambda ns} \mathbb{E}[e^{\lambda S}]. \quad (1)$$

$$= \mathbb{P}(\underbrace{e^{\lambda S}}_{>0} \geq e^{\lambda ns})$$

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可以使用 Markov's inequality

The formula above corresponds to the so-called **Chernoff's method**, which allows to bound tail probability in terms of the MGF.

Introduce an independent copy  $X'_1, \dots, X'_n$  of  $X_1, \dots, X_n$ , and note that

$$\mathbb{E}[e^{\lambda S}] = \mathbb{E}_X[e^{\lambda \sum_i (X_i - \mathbb{E}X_i)}] = \mathbb{E}_X \left[ \underbrace{\exp\left(\lambda \mathbb{E}_{X'} \sum_i (X_i - X'_i)\right)}_{\text{convex}} \right] \stackrel{\text{Jensen's inequality}}{\leq} \mathbb{E}_{X, X'} \left[ \exp\left(\lambda \sum_i (X_i - X'_i)\right) \right], \quad (2)$$

where we used Jensen's inequality with the convex function  $u \mapsto e^{\lambda u}$ .

Let  $\varepsilon_1, \dots, \varepsilon_n$  be independent random signs ( $\mathbb{P}(\varepsilon_i = \pm 1) = 1/2$ ), independent of everything else. Since  $X_i - X'_i$  is symmetric about 0, we have in distribution

$$X_i - X'_i \stackrel{d}{=} \varepsilon_i (X_i - X'_i),$$

and hence  $= \int_{-1}^1 \mathbb{E}_{X, X'} \left[ \frac{1}{2} \exp(\lambda(X_i - X'_i)) + \frac{1}{2} \exp(\lambda(X'_i - X_i)) \right] \leq \int_{-1}^1 \mathbb{E}_{X, X'} \left[ \exp\left(\frac{\lambda^2 (X_i - X'_i)^2}{2}\right) \right] \leq \exp\left(\frac{\lambda^2 n(b-a)^2}{2}\right)$

$$\mathbb{E}_{X, X'} \left[ \exp\left(\lambda \sum_i (X_i - X'_i)\right) \right] = \mathbb{E}_{X, X'} \mathbb{E}_\varepsilon \left[ \exp\left(\lambda \sum_i \varepsilon_i (X_i - X'_i)\right) \right]. \quad (3)$$

Conditional on  $X, X'$ , set  $c_i := X_i - X'_i \in [-(b-a), b-a]$ . Then

$$\mathbb{E}_\varepsilon \left[ \exp\left(\lambda \sum_i \varepsilon_i c_i\right) \right] = \prod_{i=1}^n \mathbb{E}_{\varepsilon_i} [e^{\lambda \varepsilon_i c_i}] = \prod_{i=1}^n \cosh(\lambda c_i) \leq \prod_{i=1}^n \exp\left(\frac{\lambda^2 c_i^2}{2}\right) = \exp\left(\frac{\lambda^2}{2} \sum_{i=1}^n c_i^2\right),$$

using  $\cosh u \leq e^{u^2/2}$  for all  $u \in \mathbf{R}$ . Since  $c_i^2 \leq (b-a)^2$ , this yields

$$\mathbb{E}_\varepsilon \left[ \exp\left(\lambda \sum_i \varepsilon_i (X_i - X'_i)\right) \right] \leq \exp\left(\frac{\lambda^2}{2} n (b-a)^2\right). \quad (4)$$

Combining (2)–(4) gives the mgf bound

$$\mathbb{E}[e^{\lambda S}] \leq \exp\left(\frac{\lambda^2}{2} n (b-a)^2\right). \quad (5)$$

Plug (5) into (1):

$$\mathbb{P}(\bar{X}_n - \mu \geq s) \leq \exp\left(-\lambda n s + \frac{\lambda^2}{2} n (b-a)^2\right).$$

Optimizing the right-hand side in  $\lambda$  (choose  $\lambda^* = \frac{s}{(b-a)^2}$ ) yields (希望找到最 tight 的 bound)

$$\mathbb{P}(\bar{X}_n - \mu \geq s) \leq \exp\left(-\frac{n s^2}{2(b-a)^2}\right).$$

Apply the same bound to  $-(X_i - \mu_i)$  and union bound for the two tails to obtain the two-sided version.

Confidence interval: Hoeffding inequality v.s. Chebyshev inequality

令  $X_1, \dots, X_n \in [a, b]$ ,  $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$ , 希望求  $\mu$  的 confidence interval

• Hoeffding inequality

$$P(|\bar{X} - \mu| \geq t) \leq 2 \exp\left(-\frac{2nt^2}{(b-a)^2}\right)$$

$$\Rightarrow P(|\bar{X} - \mu| \leq (b-a) \sqrt{\frac{\log(2/\delta)}{2n}}) \geq 1 - \delta$$

$$\Rightarrow 1 - \delta \text{ CI for } \mu: \bar{X} \pm (b-a) \sqrt{\frac{\log(2/\delta)}{2n}}$$

• Chebyshev inequality:

$$P(|\bar{X} - \mu| \geq t) \leq \frac{\sigma^2}{nt^2}$$

$$\Rightarrow P(|\bar{X} - \mu| \leq \sigma \sqrt{\frac{1}{n\delta}}) \geq 1 - \delta$$

$$\Rightarrow 1 - \delta \text{ CI for } \mu: \bar{X} \pm \sigma \sqrt{\frac{1}{n\delta}}$$

• Comparison:

① convergence rate 均为  $n^{-\frac{1}{2}}$

② 当  $\delta$  很小时,  $\sqrt{\frac{1}{\delta}} > \sqrt{\log(\frac{2}{\delta})}$ , 因此 Hoeffding 更好

③ 对于  $X \in [a, b]$ , 有

$$\text{Var}(X) \leq \frac{(b-a)^2}{4} \Rightarrow \sigma \leq \frac{b-a}{2} < \frac{b-a}{\sqrt{2}}$$

若  $\sigma < \frac{b-a}{\sqrt{2}}$ , 则 Chebyshev 可能更好 (Hoeffding is not sensitive to  $\text{Var}(X)$ )

e.g. 对于  $\text{Bin}(n, p)$ , Hoeffding is sharp when  $p \approx \frac{1}{2}$ , not sharp when  $p(1-p)$  is small



## 10 Lecture 11

### Central Limit Theorem vs Hoeffding: a comparison

**Central Limit Theorem (i.i.d. version).** Let  $X_1, \dots, X_n$  be i.i.d. with mean  $\mu$  and variance  $\sigma^2 \in (0, \infty)$ , and set

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i.$$

Then, for every  $a < b$ ,

$$\lim_{n \rightarrow \infty} \mathbb{P} \left\{ a \leq \frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \leq b \right\} = \int_a^b \frac{1}{\sqrt{2\pi}} e^{-z^2/2} dz.$$

Equivalently,  $\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \Rightarrow N(0, 1)$ .

Let  $X_1, \dots, X_n$  be i.i.d. with mean  $\mu$  and variance  $\sigma^2 \in (0, \infty)$ , and  $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$ . Set

$$x_n = \frac{t\sqrt{n}}{\sigma} \quad (t > 0).$$

The CLT suggests

$$\mathbb{P}(|\bar{X}_n - \mu| \geq t) \approx 2\Phi(-x_n),$$

and the Gaussian integral,

$$\left(\frac{1}{x} - \frac{1}{x^3}\right)\phi(x) \leq \Phi(-x) \leq \frac{1}{x}\phi(x) \quad (x \geq 0),$$

turn this into a more precise asymptotic with the correct *prefactor*:

$$\frac{2\sigma}{t\sqrt{2\pi n}} \exp\left(-\frac{nt^2}{2\sigma^2}\right) \left(1 - \frac{\sigma^2}{nt^2}\right) \lesssim \mathbb{P}(|\bar{X}_n - \mu| \geq t) \lesssim \frac{2\sigma}{t\sqrt{2\pi n}} \exp\left(-\frac{nt^2}{2\sigma^2}\right),$$

where  $\lesssim$  means the bounds are tight up to terms vanishing as  $n \rightarrow \infty$  (for fixed  $t$ ). Equivalently, for the one-sided tail,

$$\mathbb{P}(\bar{X}_n - \mu \geq t) \approx \Phi(-x_n) \sim \frac{\sigma}{t\sqrt{2\pi n}} \exp\left(-\frac{nt^2}{2\sigma^2}\right).$$

**Comparison with Hoeffding (bounded case).** If  $a \leq X_i \leq b$  a.s., then Hoeffding gives the finite-sample bound, for all  $n$ ,

$$\mathbb{P}(|\bar{X}_n - \mu| \geq t) \leq 2 \exp\left(-\frac{2nt^2}{(b-a)^2}\right).$$

Since  $\sigma^2 \leq (b-a)^2/4$ , the Hoeffding exponent is never larger than the CLT exponent  $-\frac{nt^2}{2\sigma^2}$ , and it lacks the Gaussian  $1/\sqrt{n}$  prefactor.

### Confidence bounds and sample size from Hoeffding

A simple practical corollary of Hoeffding's inequality is an  $(1-\delta)$  nonasymptotic confidence interval for the mean. If  $X_i \in [a, b]$  are independent and  $\bar{X}_n = \frac{1}{n} \sum_i X_i$ , then

$$\mathbb{P}\left(|\bar{X}_n - \mu| \leq (b-a) \sqrt{\frac{\log(2/\delta)}{2n}}\right) \geq 1 - \delta.$$

$\mathbb{P}(|\bar{X}_n - \mu| \geq t) \leq 2 \exp\left(-\frac{2nt^2}{(b-a)^2}\right)$   
 $\text{令 } 2 \exp\left(-\frac{2nt^2}{(b-a)^2}\right) = \delta$   
 $\text{得 } t = (b-a) \sqrt{\frac{\log(2/\delta)}{2n}}$

Equivalently, to guarantee  $|\bar{X}_n - \mu| \leq \varepsilon$  with probability  $\geq 1 - \delta$ , it suffices to take

$$n \geq \frac{(b-a)^2}{2\varepsilon^2} \log \frac{2}{\delta} \quad \leftarrow (b-a) \sqrt{\frac{\log(2/\delta)}{2n}} \leq \varepsilon$$

**Example 19** (How many coin flips?). For  $X_i \in \{0, 1\}$  (so  $b-a = 1$ ), to ensure  $|\bar{X}_n - p| \leq 0.05$  with 95% confidence,

$$n \geq \frac{1}{2(0.05)^2} \log \frac{2}{0.05} \approx 738.$$

**Example 20** (Chebyshev sample size for the same accuracy/confidence). Chebyshev gives, for  $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$  with  $\text{Var}(X_i) = p(1-p)$ ,

$$\mathbb{P}(|\bar{X}_n - p| \geq \varepsilon) \leq \frac{\text{Var}(\bar{X}_n)}{\varepsilon^2} = \frac{p(1-p)}{n\varepsilon^2}.$$

To ensure  $|\bar{X}_n - p| \leq \varepsilon$  with probability at least  $1 - \delta$ , it suffices that

$$n \geq \frac{p(1-p)}{\delta\varepsilon^2}.$$

For the same target as in the Hoeffding example ( $\varepsilon = 0.05$ ,  $\delta = 0.05$ ):

$$n \geq \frac{p(1-p)}{0.05 \cdot 0.05^2} = \frac{p(1-p)}{0.000125}.$$

- Worst case over  $p$  (unknown  $p$ ): since  $p(1-p) \leq \frac{1}{4}$ ,

$$n \geq \frac{0.25}{0.000125} = 2000.$$

- If  $p$  is small:

$$p = 0.10 \Rightarrow n \geq \frac{0.09}{0.000125} = 720, \quad p = 0.01 \Rightarrow n \geq \frac{0.0099}{0.000125} \approx 80.$$

So, Hoeffding's inequality is uniform in  $p$ . It is better than Chebyshev's inequality in the worst case  $p = \frac{1}{2}$ : it gives  $n \geq 738$  while Chebyshev needs  $n \geq 2000$ . However, Chebyshev's inequality can be tighter when  $p(1-p)$  is small (e.g.,  $n \geq 80$  at  $p = 0.01$ ).

## Empirical cdf, and the DKW inequality

**Population and empirical cdf.** Let  $X_1, \dots, X_n$  be i.i.d. copies of a r.v.  $X$  with cdf  $F(x) = \mathbb{P}(X \leq x)$ . The *empirical cdf* is

$$F_n(x) = \frac{1}{n} \sum_{i=1}^n \mathbf{1}(X_i \leq x), \quad x \in \mathbf{R}.$$

For a fixed  $x$ ,  $F_n(x)$  is an average of i.i.d. Bernoulli( $F(x)$ ) variables, hence

$$\mathbb{E}[F_n(x)] = F(x), \quad \text{Var}(F_n(x)) = \frac{F(x)(1-F(x))}{n}.$$

We can apply Hoeffding's inequality at a single point  $x$ . Since each indicator lies in  $[0, 1]$ , Hoeffding's inequality gives for any  $\varepsilon > 0$ ,

$$\mathbb{P}(|F_n(x) - F(x)| > \varepsilon) \leq 2 \exp(-2n\varepsilon^2).$$

This is a *pointwise bound* (for one fixed  $x$ ).

The Dvoretzky–Kiefer–Wolfowitz inequality strengthens the pointwise bound to a uniform one holding simultaneously for all  $x \in \mathbf{R}$ :

**Dvoretzky–Kiefer–Wolfowitz (DKW) inequality.** If  $X_1, \dots, X_n$  are i.i.d. with cdf  $F$  and empirical cdf  $F_n$ , then for every  $\varepsilon > 0$ ,

$$\mathbb{P}\left(\sup_{x \in \mathbf{R}} |F_n(x) - F(x)| > \varepsilon\right) \leq 2 \exp(-2n\varepsilon^2).$$

Equivalently, with probability at least  $1 - \delta$ ,

$$\sup_{x \in \mathbf{R}} |F_n(x) - F(x)| \leq \sqrt{\frac{1}{2n} \log \frac{2}{\delta}}.$$

**Example 21** (Distribution-free cdf confidence band). Let  $\varepsilon_n = \sqrt{\frac{1}{2n} \log \frac{2}{\delta}}$ . Then with probability  $\geq 1 - \delta$ ,

$$F_n(x) - \varepsilon_n \leq F(x) \leq F_n(x) + \varepsilon_n \quad \text{simultaneously for all } x.$$

Let  $\delta = 2 \exp(-2n\varepsilon_n^2)$   
 $\Rightarrow \varepsilon_n = \sqrt{\frac{1}{2n} \log \frac{2}{\delta}}$

# 11 Lecture 12

## 11.1 Transformation of random variables

### Change of variables (one-to-one case)

Let  $X$  be continuous with density  $f_X$ . Take  $Y = g(X)$  where  $g$  is strictly monotone on the range of  $X$  and  $g^{-1}$  exists and is differentiable there.

CDF step. If  $g$  is increasing, then

$$F_Y(y) = \mathbb{P}(Y \leq y) = \mathbb{P}(X \leq g^{-1}(y)) = F_X(g^{-1}(y)). \quad F_Y'(y) = f_X(g^{-1}(y)) \cdot (g^{-1})'(y) > 0$$

If  $g$  is decreasing, then

$$F_Y(y) = \mathbb{P}(Y \leq y) = \mathbb{P}(X \geq g^{-1}(y)) = 1 - F_X(g^{-1}(y)). \quad F_Y'(y) = -f_X(g^{-1}(y)) \cdot (g^{-1})'(y) < 0$$

Density step (chain rule). Both cases give the same compact formula:

$$f_Y(y) = f_X(g^{-1}(y)) \left| (g^{-1})'(y) \right|.$$

**Example 22** ( $X^2$  when  $X \sim N(0, 1)$ ). Here  $g(x) = x^2$  is not globally one-to-one, but it is monotone on  $(-\infty, 0]$  and  $[0, \infty)$ . For  $y > 0$ , the preimages are  $\pm\sqrt{y}$  and  $(g^{-1})'(y) = \pm\frac{1}{2\sqrt{y}}$ .

With  $\phi(x) = \frac{1}{\sqrt{2\pi}}e^{-x^2/2}$ ,

$$f_{X^2}(y) = \frac{\phi(\sqrt{y})}{2\sqrt{y}} + \frac{\phi(-\sqrt{y})}{2\sqrt{y}} = \frac{\phi(\sqrt{y})}{\sqrt{y}} = \frac{1}{\sqrt{2\pi y}} e^{-y/2}, \quad y > 0,$$

and  $f_{X^2}(y) = 0$  for  $y \leq 0$ .

### Quantiles and the inverse cdf

**Definition 14** (Quantile function). Let  $F_X$  be the cdf of a r.v.  $X$ . For  $u \in (0, 1)$  define the quantile

$$q_X(u) := \inf\{x \in \mathbf{R} : F_X(x) \geq u\}.$$

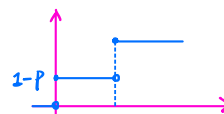
If  $F_X$  is continuous and strictly increasing, then  $q_X(u) = F_X^{-1}(u)$ .

**Example 23** (Uniform). If  $X \sim \text{Unif}(0, 1)$ , then  $F_X(x) = x$  on  $(0, 1)$ , hence  $q_X(u) = u$  for  $u \in (0, 1)$ .

**Example 24** (Normal). If  $X \sim N(0, 1)$  with cdf  $\Phi$ , then  $q_X(u) = \Phi^{-1}(u)$  (no elementary closed form).

**Example 25** (Bernoulli). If  $X \sim \text{Bernoulli}(p)$ , then for  $u \in (0, 1)$ ,

$$q_X(u) = \begin{cases} 0, & 0 < u \leq 1-p, \\ 1, & 1-p < u < 1. \end{cases} \quad F_X(x) = \begin{cases} 0, & x < 0 \\ 1-p, & 0 \leq x < 1 \\ 1, & x \geq 1 \end{cases}$$



## 12 Lecture 13

**Fact (Probability integral transform; inverse-cdf sampling).**

1. If  $U \sim \text{Unif}(0, 1)$  and  $F$  is a continuous, strictly increasing cdf, then  $F^{-1}(U)$  has cdf  $F$ .
2. If  $X$  has continuous, increasing cdf  $F$ , then  $F(X) \sim \text{Unif}(0, 1)$ .

Indeed, to show this we have for any  $x \in \mathbf{R}$ ,

$$\mathbb{P}(F^{-1}(U) \leq x) = \mathbb{P}(U \leq F(x)) = F(x),$$

since  $U$  is uniform and  $F$  is increasing. Thus the cdf of  $F^{-1}(U)$  equals  $F$ . On the other side, for any  $t \in (0, 1)$ ,

$$\mathbb{P}(F(X) \leq t) = \mathbb{P}(X \leq F^{-1}(t)) = F(F^{-1}(t)) = t,$$

so  $F(X)$  is uniform on  $(0, 1)$ .

### Joint densities

**Definition 15 (Joint density).** A function  $f : \mathbf{R}^2 \rightarrow [0, \infty)$  is a *joint density* for  $(X, Y)$  if

$$\iint_{\mathbf{R}^2} f(x, y) \, dx \, dy = 1 \quad \text{and} \quad \mathbb{P}((X, Y) \in B) = \iint_B f(x, y) \, dx \, dy$$

for every set  $B \subset \mathbf{R}^2$ . We usually write  $f_{X,Y} = f$ .

**Marginals.** They are given by the following formulas:

$$f_X(x) = \int_{-\infty}^{\infty} f(x, y) \, dy, \quad f_Y(y) = \int_{-\infty}^{\infty} f(x, y) \, dx.$$

**Fact. Independence is equivalent to factorization of the joint density.** Let  $(X, Y)$  have joint density  $f$  and marginals  $f_X, f_Y$ . Then the following are equivalent:

1.  $X$  and  $Y$  are independent.
2.  $f(x, y) = g(x)h(y)$  for some nonnegative functions  $g, h$  with  $\int g = \int h = 1$ .
3.  $f(x, y) = f_X(x) f_Y(y)$  for every  $(x, y) \in \mathbf{R}^2$ .

**Probabilities over a general set  $A$ .** For any measurable  $A \subset \mathbf{R}^2$ ,

$$\mathbb{P}((X, Y) \in A) = \iint_A f(x, y) \, dx \, dy = \int_{-\infty}^{\infty} \left( \int_{-\infty}^{\infty} \mathbf{1}((x, y) \in A) f(x, y) \, dy \right) dx.$$

If  $A = \{(x, y) : a \leq x \leq b, g_1(x) \leq y \leq g_2(x)\}$ , then

$$\mathbb{P}((X, Y) \in A) = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y) \, dy \, dx.$$

**Example 26** (Uniform on the unit square). Let

$$f(x, y) = \mathbf{1}(0 \leq x \leq 1) \mathbf{1}(0 \leq y \leq 1).$$

Then  $f_X(x) = \mathbf{1}(0 \leq x \leq 1)$ ,  $f_Y(y) = \mathbf{1}(0 \leq y \leq 1)$  and  $f(x, y) = f_X(x)f_Y(y)$  everywhere; hence  $X$  and  $Y$  are independent and each is uniformly distributed on the  $(0, 1)$  interval.

**Example 27** (Uniform on the unit disk; dependence). Let

$$f(x, y) = \frac{1}{\pi} \mathbf{1}(x^2 + y^2 \leq 1).$$

Then

$$f_X(x) = \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} \frac{1}{\pi} \, dy = \frac{2}{\pi} \sqrt{1-x^2} \mathbf{1}(|x| \leq 1), \quad f_Y(y) = \frac{2}{\pi} \sqrt{1-y^2} \mathbf{1}(|y| \leq 1).$$

At  $(x, y) = (0, 0)$  we have  $f(0, 0) = 1/\pi$  but  $f_X(0)f_Y(0) = (2/\pi)^2 \neq 1/\pi$ . Thus  $f \neq f_X f_Y$  and  $X, Y$  are *not* independent (intuitively,  $x^2 + y^2 \leq 1$  couples  $X$  and  $Y$ ).

**Expectation of a function.** For any  $h : \mathbf{R}^2 \rightarrow \mathbf{R}$  with  $\iint |h|f < \infty$ ,

$$\mathbb{E}[h(X, Y)] = \iint_{\mathbf{R}^2} h(x, y) f(x, y) \, dx \, dy.$$

## 13 Lecture 14

### Conditional densities (continuous case)

Throughout this section  $(X, Y)$  denotes a pair of real-valued random variables with joint density  $f_{X,Y}(x, y)$ . The one-dimensional (marginal) densities are obtained by integrating the joint:

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) \, dy, \quad f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) \, dx.$$

**Why the usual “ratio”**  $\mathbb{P}(A \mid B) = \mathbb{P}(AB)/\mathbb{P}(B)$  **is not usable at a point.** For a fixed number  $y$  the event  $\{Y = y\}$  has probability 0 (because  $Y$  is continuous), so the naïve expression

$$\mathbb{P}\{X \in A \mid Y = y\} = \frac{\mathbb{P}\{X \in A, Y = y\}}{\mathbb{P}\{Y = y\}}$$

is the undefined ratio  $0/0$ . The right way is to *thicken* the conditioning event to a small strip  $y - \varepsilon/2 \leq Y \leq y + \varepsilon/2$ , compute the ordinary conditional probability, and then let the thickness  $\varepsilon \downarrow 0$ :

$$\mathbb{P}\{X \in A \mid Y = y\} = \lim_{\varepsilon \downarrow 0} \frac{\int_A \int_{y-\varepsilon/2}^{y+\varepsilon/2} f_{X,Y}(x, t) \, dt \, dx}{\int_{y-\varepsilon/2}^{y+\varepsilon/2} f_Y(t) \, dt} = \int_A \frac{f_{X,Y}(x, y)}{f_Y(y)} \, dx.$$

**Definition 16** (Conditional density of  $X$  given  $Y = y$ ). For any  $y$  with  $f_Y(y) > 0$ ,

$$f_{X|Y=y}(x) := \frac{f_{X,Y}(x, y)}{f_Y(y)}.$$

Then for any interval (or finite union of intervals)  $A \subset \mathbf{R}$ ,

$$\mathbb{P}\{X \in A \mid Y = y\} = \int_A f_{X|Y=y}(x) \, dx.$$

**Proportionality to the joint.** For fixed  $y$ , the function  $x \mapsto f_{X|Y=y}(x)$  is *proportional* to the joint  $x \mapsto f_{X,Y}(x, y)$ :

$$f_{X|Y=y}(x) = \frac{1}{f_Y(y)} f_{X,Y}(x, y),$$

and the constant  $1/f_Y(y)$  is determined by the requirement that the conditional density integrates to 1 in  $x$ .

**Law of total probability (continuous version).** From  $f_{X,Y}(x, y) = f_{X|Y=y}(x) f_Y(y)$  and integrating over  $y$  we obtain

$$f_X(x) = \int_{-\infty}^{\infty} f_{X|Y=y}(x) f_Y(y) \, dy.$$

So the marginal density of  $X$  is an *average* of the conditional densities of  $X$  at different  $y$ , with weights  $f_Y(y)$ .

## Sum of two independent variables: convolution via conditional densities

Let  $X$  and  $Y$  be independent with densities  $f_X$  and  $f_Y$ , and set  $Z = X + Y$ . Condition on the value of  $Y$  and use the previous identity:

$$f_Z(z) = \int_{-\infty}^{\infty} f_{Z|Y=y}(z) f_Y(y) dy.$$

Given  $Y = y$ , the sum is  $Z = X + y$ , hence  $f_{Z|Y=y}(z) = f_{X|Y=y}(z - y)$ . Independence tells us  $f_{X|Y=y} = f_X$  for every  $y$ , therefore

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(z - y) f_Y(y) dy = \int_{-\infty}^{\infty} f_X(x) f_Y(z - x) dx.$$

$$f_{X+Y}(z) = (f_X * f_Y)(z) := \int_{-\infty}^{\infty} f_X(x) f_Y(z - x) dx.$$

This integral is the *convolution* of the two densities. Intuitively, to get  $Z = z$  we sum over all ways to split  $z$  into  $x + (z - x)$  and add the corresponding probabilities.

## Multivariate change of variables (Jacobian)

**Fact (Jacobian change of variables).** Let  $(X, Y)$  have joint density  $f_{X,Y}$  on  $\mathbf{R}^2$  and let  $(U, V) = T(X, Y)$ , where  $T : \mathbf{R}^2 \rightarrow \mathbf{R}^2$  is a one-to-one transformation with inverse  $S = T^{-1}$ . Then

$$f_{U,V}(u, v) = f_{X,Y}(S(u, v)) |\det J_S(u, v)|,$$

where  $J_S$  is the Jacobian matrix of  $S$ . This is the two-dimensional analogue of the one-variable rule  $f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|$  for a monotone transform  $Y = g(X)$ .

### 13.1 Representation of independent Gaussian random variables

We now see how *polar coordinates* give a clean description of a pair of independent centered Gaussians.

Consider the polar transform  $(R, \Theta) = T(X, Y)$  with inverse

$$S(r, \theta) = (x, y) = (r \cos \theta, r \sin \theta), \quad r > 0, \quad 0 < \theta < 2\pi.$$

The Jacobian matrix and its determinant are

$$J_S(r, \theta) = \begin{pmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{pmatrix}, \quad |\det J_S(r, \theta)| = r,$$

hence the joint density transforms as

$$f_{R,\Theta}(r, \theta) = r f_{X,Y}(r \cos \theta, r \sin \theta), \quad r > 0, \quad 0 < \theta < 2\pi.$$



• 一个极坐标转换的例子 (2 iid.  $N(0, \sigma^2)$ )

## 14 Lecture 15

**Example 28** (Independent  $N(0, \sigma^2)$  components in polar form). If  $X \sim N(0, \sigma^2)$  and  $Y \sim N(0, \sigma^2)$  are independent, then

$$f_{X,Y}(x, y) = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{x^2 + y^2}{2\sigma^2}\right).$$

Using the polar formula,

$$f_{R,\Theta}(r, \theta) = r \frac{1}{2\pi\sigma^2} \exp\left(-\frac{r^2}{2\sigma^2}\right) = \underbrace{\frac{1}{2\pi}}_{\Theta \sim \text{Unif}(0, 2\pi)} \cdot \underbrace{\frac{r}{\sigma^2} e^{-r^2/(2\sigma^2)}}_{\text{Rayleigh}(\sigma)},$$

for  $r > 0$ ,  $0 < \theta < 2\pi$ . Thus  $R$  and  $\Theta$  are independent,

$$R \sim \text{Rayleigh}(\sigma), \quad \Theta \sim \text{Unif}(0, 2\pi),$$

and  $R^2 = X^2 + Y^2$  in distribution (Check this!).

**Converse construction.** Let  $\Theta \sim \text{Unif}(0, 2\pi)$  and let  $R$  be independent of  $\Theta$  with

$$f_R(r) = \frac{r}{\sigma^2} e^{-r^2/(2\sigma^2)}, \quad r > 0.$$

Define

$$X = R \cos \Theta, \quad Y = R \sin \Theta.$$

Then  $X$  and  $Y$  are independent  $N(0, \sigma^2)$ .

*Proof.* The joint density of  $(R, \Theta)$  is

$$f_{R,\Theta}(r, \theta) = f_R(r) f_\Theta(\theta) = \frac{1}{2\pi} \frac{r}{\sigma^2} e^{-r^2/(2\sigma^2)}, \quad r > 0, \quad 0 < \theta < 2\pi.$$

Use the bijection  $(r, \theta) \mapsto (x, y) = (r \cos \theta, r \sin \theta)$ . Its Jacobian determinant is  $r$ , so the absolute determinant for the inverse map is  $1/r$ . By the Jacobian formula,

$$\begin{aligned} f_{X,Y}(x, y) &= f_{R,\Theta}(\sqrt{x^2 + y^2}, \theta(y, x)) \frac{1}{\sqrt{x^2 + y^2}} \quad \text{det}(A^{-1}) = \frac{1}{\text{det}(A)} \\ &= \frac{1}{2\pi} \frac{\sqrt{x^2 + y^2}}{\sigma^2} e^{-(x^2 + y^2)/(2\sigma^2)} \cdot \frac{1}{\sqrt{x^2 + y^2}} \\ &= \frac{1}{2\pi\sigma^2} \exp\left(-\frac{x^2 + y^2}{2\sigma^2}\right). \end{aligned}$$

Integrating out  $y$  gives, for every  $x$ ,

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dy = \frac{1}{\sqrt{2\pi}\sigma} e^{-x^2/(2\sigma^2)} = \phi_{\sigma}(x),$$

and similarly  $f_Y(y) = \phi_{\sigma}(y)$  with  $\phi_{\sigma}(t) = \frac{1}{\sqrt{2\pi}\sigma} e^{-t^2/(2\sigma^2)}$ . Finally,

$$f_{X,Y}(x, y) = \phi_{\sigma}(x) \phi_{\sigma}(y) = f_X(x) f_Y(y) \quad (\text{all } x, y),$$

so the joint factorizes into the product of its marginals; therefore  $X$  and  $Y$  are independent and each is  $N(0, \sigma^2)$ .

• Conditional Exp.  
• Law of total Exp.

## 15 Lecture 16

### Conditional expectation as a function

Throughout,  $(X, Y)$  denotes a pair of real random variables. When  $(X, Y)$  has a joint density  $f_{X,Y}$  and  $f_X(x) > 0$ , the **conditional density** of  $Y$  given  $X = x$  is

$$f_{Y|X=x}(y) = \frac{f_{X,Y}(x, y)}{f_X(x)}.$$

$X, Y$  independent

$$\iff f_{X,Y}(x, y) = f_X(x) \cdot f_Y(y)$$

$$\iff f_{Y|X=x}(y) = f_Y(y), \quad \forall x$$

The **conditional expectation** (also called the *regression function* of  $Y$  on  $X$ ) is

$X, Y$  independent

$$\mathbb{E}[Y | X = x] = \int_{-\infty}^{\infty} y f_{Y|X=x}(y) dy = \frac{\int_{-\infty}^{\infty} y f_{X,Y}(x, y) dy}{f_X(x)}.$$

$$\iff f_{X,Y}(x, y) = f_X(x) \cdot f_Y(y)$$

$$\iff \mathbb{E}[Y|X=x] = \mathbb{E}[Y], \quad \forall x$$

More generally, for any function  $g$  for which the integrals exist,

$$\mathbb{E}[g(Y) | X = x] = \int g(y) f_{Y|X=x}(y) dy,$$

$$\mathbb{E}[g(X, Y) | X = x] = \int g(x, y) f_{Y|X=x}(y) dy.$$

**Conditional expectation as a random variable.** The map  $x \mapsto m(x) = \mathbb{E}[Y | X = x]$  induces a new random variable  $m(X)$ , which we denote by

$$\mathbb{E}[Y | X] = m(X).$$

Informally,  $\mathbb{E}[Y | X]$  is the “best”  $X$ -based prediction of  $Y$  under squared loss; we will return to this below.

**Law of total expectation (tower property).** Averaging the regression function brings us back to the global mean:

$$\mathbb{E}[Y] = \mathbb{E}[\mathbb{E}[Y | X]].$$

*Proof (continuous case).* Using  $f_Y(y) = \int f_{X,Y}(x, y) dx$  and changing the order of integration,

$$\begin{aligned} \mathbb{E}[Y] &= \int y f_Y(y) dy = \int y \left( \int f_{X,Y}(x, y) dx \right) dy = \int \left( \int y f_{Y|X=x}(y) dy \right) f_X(x) dx \\ &= \int \mathbb{E}[Y | X = x] f_X(x) dx = \mathbb{E}[\mathbb{E}[Y | X]]. \end{aligned}$$

### Quick examples via the tower property

**Example 29** (Two uniform cuts on a stick). Consider a stick of length  $\ell$ . Break it at a random point  $X$  that is chosen uniformly across the length of the stick. Then break the stick again at a random point  $Y$  that is also chosen uniformly across the length of the stick. What is the expected length of the final piece?

Pick the first cut  $X \sim \text{Unif}(0, \ell)$ , so  $f_X(x) = \frac{1}{\ell}$  on  $(0, \ell)$ . Given  $X = x$ , place the second cut uniformly on  $(0, x)$ , i.e.,

$$f_{Y|X=x}(y) = \frac{1}{x} \text{ for } 0 < y < x \text{ (and 0 otherwise).}$$

Then

$$\mathbb{E}[Y | X = x] = \int_0^x y \frac{1}{x} dy = \frac{1}{x} \cdot \frac{x^2}{2} = \frac{x}{2}.$$

Average over  $X$ :

$$\mathbb{E}[Y] = \mathbb{E}[\mathbb{E}[Y | X]] = \int_0^\ell \frac{x}{2} \frac{1}{\ell} dx = \frac{1}{2\ell} \cdot \frac{\ell^2}{2} = \frac{\ell}{4}.$$

**Example 30** ( $\mathbb{P}\{X \leq YZ\}$  with  $X, Y, Z$  i.i.d.  $\text{Unif}(0, 1)$ ). Suppose  $X, Y, Z$  are i.i.d  $\text{Unif}(0, 1)$  random variables. Find the value of  $\mathbb{P}\{X \leq YZ\}$ .

Condition on  $(Y, Z) = (y, z)$ . Since  $X$  is independent of  $(Y, Z)$  and uniform on  $(0, 1)$ ,

$$\mathbb{P}\{X \leq yz | Y = y, Z = z\} = \mathbb{P}\{X \leq yz\} = yz \quad (0 < y, z < 1).$$

Therefore, by the tower property, 即使  $X, Y, Z$  i.i.d.

$$\mathbb{P}\{X \leq YZ\} = \mathbb{E}[\mathbb{P}\{X \leq YZ | Y, Z\}] = \mathbb{E}[YZ] = \left( \int_0^1 y dy \right) \left( \int_0^1 z dz \right) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}.$$

**Example 31** (Wald's identity). Assume  $\mathbb{E}|X_1| < \infty$ ,  $\mathbb{E}[N] < \infty$ , and  $N$  is independent of the i.i.d. sequence  $X_1, X_2, \dots$  with  $\mathbb{E}[X_1] = \mu$ . For  $S = \sum_{i=1}^N X_i$ ,

$$\mathbb{E}[S | N = n] = \mathbb{E}\left[\sum_{i=1}^n X_i\right] = n\mu \Rightarrow \mathbb{E}[S | N] = N\mu.$$

Taking expectations,

$$\mathbb{E}[S] = \mathbb{E}[\mathbb{E}[S | N]] = \mathbb{E}[N\mu] = \mu \mathbb{E}[N].$$

- Squared loss predictor
- Abs. loss predictor
- 0-1 loss predictor

## 16 Lecture 17

### Best predictor (risk minimization)

**Squared loss.** Let  $g : \mathbf{R} \rightarrow \mathbf{R}$  be any prediction rule and write

$$R(g) := \mathbb{E}[(Y - g(X))^2]. \quad (\text{condition on } (X, Y))$$

Set the regression function  $m(X) := \mathbb{E}[Y | X]$ . Using the tower property and conditioning on  $X$ , *对x求 对y求*

$$\begin{aligned} R(g) &= \mathbb{E}[\mathbb{E}[(Y - g(X))^2 | X]] \quad (\text{使用 tower property 能不用考虑联合分布}) \\ &= \mathbb{E}[\mathbb{E}[(Y - m(X) + m(X) - g(X))^2 | X]] \\ &= \mathbb{E}[\mathbb{E}[(Y - m(X))^2 | X] + 2(m(X) - g(X)) \underbrace{\mathbb{E}[Y - m(X) | X]}_{=0} + (m(X) - g(X))^2] \\ &= \mathbb{E}[\text{Var}(Y | X)] + \mathbb{E}[(m(X) - g(X))^2]. \end{aligned}$$

The first term does *not* depend on  $g$ . The second term is minimized iff  $g(X) = m(X)$ . Therefore,

$$g^*(x) = \mathbb{E}[Y | X = x] \quad \text{and} \quad R(g) = \mathbb{E}[\text{Var}(Y | X)] + \mathbb{E}[(m(X) - g(X))^2].$$

**Absolute loss (full derivation in the continuous case).** Let *若对y given x, c(x) 均 minimizes  $\phi_x(c)$  则 c(x) minimizes  $R(g)$ .*

$$R(g) := \mathbb{E}[|Y - g(X)|], \quad \phi_x(c) = \mathbb{E}[|Y - c| | X = x] = \int_{-\infty}^{\infty} |y - c| f_{Y|X=x}(y) dy.$$

Assume for simplicity that  $Y | X = x$  has a continuous density. Differentiate under the integral:

$$\phi'_x(c) = \int_{-\infty}^{\infty} \frac{\partial}{\partial c} |y - c| f_{Y|X=x}(y) dy = \int_{-\infty}^c 1 \cdot f_{Y|X=x}(y) dy - \int_c^{\infty} 1 \cdot f_{Y|X=x}(y) dy = 2F_{Y|X=x}(c) - 1.$$

Hence any minimizer  $c^*$  of  $\phi_x$  satisfies  $F_{Y|X=x}(c^*) = \frac{1}{2}$ , i.e.  $c^*$  is a *median* of  $Y | X = x$ . Therefore, one may take  $g^*(x) = \text{median}(Y | X = x)$ .

**0-1 loss (classification).** Let  $Y \in \{0, 1\}$  and let  $g : \mathbf{R}^d \rightarrow \{0, 1\}$  be any decision rule. The misclassification risk can be written by conditioning on  $X$ :

$$R(g) = \mathbb{P}\{g(X) \neq Y\} = \mathbb{E}[\mathbb{P}\{g(X) \neq Y | X\}].$$

Define the regression function

*(同样的策略 先 condition on X)*

$$\eta(x) = \mathbb{P}(Y = 1 | X = x).$$

For a fixed  $x$ , the *conditional* error of  $g$  is

$$\begin{aligned} r_x(g) &:= \mathbb{P}\{g(X) \neq Y \mid X = x\} = 1 - \mathbb{P}\{Y = g(x) \mid X = x\} \\ &= 1 - \left( \mathbf{1}(g(x) = 1) \mathbb{P}\{Y = 1 \mid X = x\} + \mathbf{1}(g(x) = 0) \mathbb{P}\{Y = 0 \mid X = x\} \right) \quad (\star) \\ &= 1 - \left( \mathbf{1}(g(x) = 1) \eta(x) + \mathbf{1}(g(x) = 0) (1 - \eta(x)) \right). \quad (\text{改为 risk function}) \end{aligned}$$

The *Bayes rule* is

$$g^*(x) = \mathbf{1}(\eta(x) \geq \frac{1}{2}). \quad \leftarrow \text{证明}$$

Write  $I_g(x) = \mathbf{1}(g(x) = 1)$  and  $I_{g^*}(x) = \mathbf{1}(g^*(x) = 1)$ . Subtract the conditional risks of  $g$  and  $g^*$  using  $(\star)$ :

$$\begin{aligned} r_x(g) - r_x(g^*) &= (I_{g^*}(x) - I_g(x)) \eta(x) + ((1 - I_{g^*}(x)) - (1 - I_g(x))) (1 - \eta(x)) \\ &= (I_{g^*}(x) - I_g(x)) (\eta(x) - (1 - \eta(x))) \quad (\dagger) \\ &= (2\eta(x) - 1) (I_{g^*}(x) - I_g(x)). \quad = \begin{cases} 1 - I_g(x) > 0 & \text{若 } \eta(x) > \frac{1}{2} \\ 0 - I_g(x) \leq 0 & \text{若 } \eta(x) \leq \frac{1}{2} \end{cases} \end{aligned}$$

*Sign check.* If  $\eta(x) > \frac{1}{2}$ , then  $2\eta(x) - 1 > 0$  and  $g^*(x) = 1$ , so  $I_{g^*}(x) - I_g(x) \in \{0, 1\} \geq 0$ , hence  $r_x(g) - r_x(g^*) \geq 0$ . If  $\eta(x) < \frac{1}{2}$ , then  $2\eta(x) - 1 < 0$  and  $g^*(x) = 0$ , so  $I_{g^*}(x) - I_g(x) \in \{0, -1\} \leq 0$ , again  $r_x(g) - r_x(g^*) \geq 0$ . If  $\eta(x) = \frac{1}{2}$ , both labels are optimal and  $(\dagger) = 0$ .

Therefore  $r_x(g) \geq r_x(g^*)$  for every  $x$ , and averaging over  $X$  gives

$$R(g) = \mathbb{E}[r_X(g)] \geq \mathbb{E}[r_X(g^*)] = R(g^*).$$

Thus  $g^*$  minimizes the 0-1 risk. Moreover,

$$r_x(g^*) = \min\{\eta(x), 1 - \eta(x)\}, \quad R(g^*) = \mathbb{E}[\min\{\eta(X), 1 - \eta(X)\}],$$

the *Bayes error*.

## 17 Lecture 18

### Conditional variance and the law of total variance

**Definition (density case).** For  $x$  with  $f_X(x) > 0$  and  $m(x) = \mathbb{E}[Y \mid X = x]$ ,

$$\text{Var}(Y \mid X = x) = \mathbb{E}[(Y - m(x))^2 \mid X = x] = \int_{-\infty}^{\infty} (y - m(x))^2 f_{Y|X=x}(y) dy = \mathbb{E}[Y^2 \mid X = x] - m(x)^2.$$

The random variable  $\text{Var}(Y \mid X)$  is obtained by evaluating the expression at  $X$ .

**Law of total variance.** Let  $m(X) = \mathbb{E}[Y | X]$ . Then

$$\begin{aligned}\text{Var}(Y) &= \mathbb{E}[(Y - \mathbb{E}Y)^2] = \mathbb{E}[(Y - m(X) + m(X) - \mathbb{E}Y)^2] \\ &= \mathbb{E}[(Y - m(X))^2] + 2 \mathbb{E}[(Y - m(X))(m(X) - \mathbb{E}Y)] + \mathbb{E}[(m(X) - \mathbb{E}Y)^2].\end{aligned}$$

The cross term vanishes:

$$\mathbb{E}[(Y - m(X))(m(X) - \mathbb{E}Y)] \stackrel{\text{tower property (condition on } X)}{=} \mathbb{E}\left[\underbrace{\mathbb{E}[Y - m(X) | X]}_{=0} \underbrace{(m(X) - \mathbb{E}Y)}_{\substack{= \mathbb{E}[\mathbb{E}[Y|X]] \\ = \mathbb{E}[m(X)]}}\right] = 0.$$

仅关于X, 在condition on X后可被提出

For the first term, apply the tower property

$$\mathbb{E}[(Y - m(X))^2] = \mathbb{E}[\mathbb{E}[(Y - m(X))^2 | X]] \stackrel{\text{定义}}{=} \mathbb{E}[\text{Var}(Y | X)].$$

The last term equals  $\text{Var}(m(X)) = \text{Var}(\mathbb{E}[Y | X])$ . Hence

$$\text{Var}(Y) = \mathbb{E}[\text{Var}(Y | X)] + \text{Var}(\mathbb{E}[Y | X]).$$

**Example 32** (Random sum: expectation and variance). Let  $X_1, X_2, \dots$  be i.i.d. with  $\mathbb{E}[X_1] = \mu$  and  $\text{Var}(X_1) = \sigma^2 < \infty$ . Let  $N$  be an  $\{1, 2, \dots\}$ -valued random variable independent of the sequence, and define

$$S = \sum_{i=1}^N X_i.$$

As we have shown above:

$$\mathbb{E}[S | N = n] = n\mu \quad \Rightarrow \quad \mathbb{E}[S] = \mathbb{E}[\mathbb{E}[S | N]] = \mu \mathbb{E}[N].$$

For the variance, we have

$$\text{Var}(S | N = n) = \text{Var}\left(\sum_{i=1}^n X_i\right) = n\sigma^2 \quad \Rightarrow \quad \mathbb{E}[\text{Var}(S | N)] = \mathbb{E}[N\sigma^2] = \sigma^2 \mathbb{E}[N].$$

Second,

$$\mathbb{E}[S | N] = N\mu \quad \Rightarrow \quad \text{Var}(\mathbb{E}[S | N]) = \text{Var}(N\mu) = \mu^2 \text{Var}(N).$$

Putting the two pieces together,

$$\text{Var}(S) = \mathbb{E}[\text{Var}(S | N)] + \text{Var}(\mathbb{E}[S | N]) = \sigma^2 \mathbb{E}[N] + \mu^2 \text{Var}(N).$$

## Random vectors, expectation, and covariance

A **random vector** is a column vector whose entries are random variables. Write

$$Y = \begin{pmatrix} Y_1 \\ \vdots \\ Y_p \end{pmatrix} \in \mathbf{R}^p.$$

Its **expectation** is the vector of coordinate-wise means

$$\mathbb{E}[Y] = \begin{pmatrix} \mathbb{E}[Y_1] \\ \vdots \\ \mathbb{E}[Y_p] \end{pmatrix}.$$

The **covariance matrix** of  $Y$  is

$$\text{Cov}(Y) = \mathbb{E}[(Y - \mathbb{E}Y)(Y - \mathbb{E}Y)^\top] \in \mathbf{R}^{p \times p}.$$

Equivalently, the  $(i, j)$  entry is  $\text{Cov}(Y_i, Y_j) = \mathbb{E}[(Y_i - \mathbb{E}Y_i)(Y_j - \mathbb{E}Y_j)]$ .

### Basic facts:

1. The diagonal entries of  $\text{Cov}(Y)$  are the variances:  $(\text{Cov}(Y))_{ii} = \text{Var}(Y_i)$ .
2.  $\text{Cov}(Y)$  is symmetric and positive semidefinite:  $\text{Cov}(Y) = \text{Cov}(Y)^\top$  and  $a^\top \text{Cov}(Y) a \geq 0$  for all  $a \in \mathbf{R}^p$ .

**Linear transforms.** Let  $A$  be a deterministic  $m \times p$  matrix and  $c \in \mathbf{R}^m$ . For  $Z = AY + c$ ,

$$\mathbb{E}[Z] = A \mathbb{E}[Y] + c, \quad \text{Cov}(Z) = A \text{Cov}(Y) A^\top.$$

In particular, for a vector  $a \in \mathbf{R}^p$ ,

$$\text{Var}(a^\top Y) = a^\top \text{Cov}(Y) a = \sum_{i,j=1}^p a_i a_j \text{Cov}(Y_i, Y_j).$$

## The multivariate normal family

**Building  $N_p(0, I_p)$  from independent Gaussians.** Start with  $p$  independent standard normals

$$Z_1, \dots, Z_p \text{ i.i.d. } \sim N(0, 1), \quad Z = \begin{pmatrix} Z_1 \\ \vdots \\ Z_p \end{pmatrix} \in \mathbf{R}^p.$$

*$n$  个 independent  $N(0,1)$  可以构造  $MVN(0, I_p)$*

By independence, the joint density is the product of the univariate densities:

$$\begin{aligned} f_Z(z) &= \prod_{i=1}^p \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}z_i^2\right) = \frac{1}{(2\pi)^{p/2}} \exp\left(-\frac{1}{2}\sum_{i=1}^p z_i^2\right) \\ &= \frac{1}{(2\pi)^{p/2}} \exp\left(-\frac{1}{2}z^\top z\right), \quad z = (z_1, \dots, z_p)^\top \in \mathbf{R}^p. \end{aligned}$$

From the construction,

$$\mathbb{E}[Z] = 0, \quad \text{Cov}(Z) = I_p,$$

since  $\text{Var}(Z_i) = 1$  and  $\text{Cov}(Z_i, Z_j) = 0$  for  $i \neq j$  (independence). We therefore write  $Z \sim N_p(0, I_p)$ .

## 18 Lecture 19

### From independent Gaussians to a general $N_p(\mu, \Sigma)$ .

A convenient way to build a multivariate normal is to start from independent standard normals and apply a linear map followed by a shift. Fix  $\mu \in \mathbf{R}^p$  and an invertible  $p \times p$  matrix  $A$ . Define the affine transform

$$X = \mu + AZ, \quad Z \sim N_p(0, I_p).$$

Because  $\mathbb{E}[Z] = 0$  and  $\text{Cov}(Z) = I_p$ , we immediately get

$$\mathbb{E}[X] = \mu + A\mathbb{E}[Z] = \mu, \quad \text{Cov}(X) = A \underbrace{\text{Cov}(Z)}_{=I_p} A^\top = AA^\top.$$

Let  $\Sigma = AA^\top$ ; this matrix is symmetric and positive definite. To write the density of  $X$ , use the Jacobian formula with the inverse map  $(x \mapsto z) = A^{-1}(x - \mu)$ :

$$\begin{aligned} f_X(x) &= f_Z(A^{-1}(x - \mu)) |\det(A^{-1})| \\ &= \frac{1}{(2\pi)^{p/2}} \exp\left(-\frac{1}{2}(x - \mu)^\top (A^{-1})^\top A^{-1}(x - \mu)\right) \frac{1}{|\det A|}. \end{aligned}$$

Since  $(A^{-1})^\top A^{-1} = \Sigma^{-1}$  and  $|\det A| = \sqrt{\det \Sigma}$ , this becomes  $(\det(\Sigma) = \det(AA^\top) = \det(A)\det(A^\top) = \det(A)^2)$

$$f_X(x) = \frac{1}{(2\pi)^{p/2} \sqrt{\det \Sigma}} \exp\left(-\frac{1}{2}(x - \mu)^\top \Sigma^{-1}(x - \mu)\right), \quad x \in \mathbf{R}^p.$$

We write  $X \sim N_p(\mu, \Sigma)$ . Conversely, any  $N_p(\mu, \Sigma)$  can be generated as  $X = \mu + AZ$  with  $Z \sim N_p(0, I_p)$  and any  $A$  satisfying  $AA^\top = \Sigma$ . This construction is useful to remember: *linear image of independent standard Gaussians + shift.*

**Stability under linear maps.** If  $X \sim N_p(\mu, \Sigma)$  and  $B$  is an  $m \times p$  matrix,  $b \in \mathbf{R}^m$ , then

$$BX + b \sim N_m(B\mu + b, B\Sigma B^\top).$$

In particular, every one-dimensional projection is normal: for any  $a \in \mathbf{R}^p$ ,

$$a^\top X \sim N(a^\top \mu, a^\top \Sigma a).$$

This shows that the normal family is closed under taking linear combinations.



若  $\Sigma$  diagonal,  $\mathbb{R}^p$   $\Sigma^{-1} = \begin{bmatrix} 1/\sigma_1^2 & & 0 \\ & \ddots & \\ 0 & & 1/\sigma_p^2 \end{bmatrix}$

$$\Rightarrow \begin{cases} (x-\mu)^T \Sigma^{-1} (x-\mu) = \sum_{i=1}^p (x_i - \mu_i)^2 / \sigma_i^2 \\ \det(\Sigma) = \prod_{i=1}^p \sigma_i^2 \end{cases} \Rightarrow f_X(x) = \frac{1}{(2\pi)^{p/2} \sqrt{\det(\Sigma)}} \exp\left\{-\frac{1}{2} \frac{(x-\mu)^T \Sigma^{-1} (x-\mu)}{\det(\Sigma)}\right\} = \prod_{i=1}^p \frac{1}{(2\pi)^{1/2} \sigma_i} \exp\left\{-\frac{(x_i - \mu_i)^2}{2\sigma_i^2}\right\} \Rightarrow X_i \text{ independent}$$

**Independence of coordinates.** If  $X \sim N_p(\mu, \Sigma)$  and  $\Sigma$  is diagonal, then the coordinates  $X_1, \dots, X_p$  are independent with  $X_i \sim N(\mu_i, \Sigma_{ii})$ . For multivariate normals, lack of correlation is as strong as independence:

**Fact (Gaussian: uncorrelated  $\Leftrightarrow$  independent).** If  $X \sim N_p(\mu, \Sigma)$ , then  $X_i$  and  $X_j$  are independent if and only if  $\text{Cov}(X_i, X_j) = 0$ . ✱ 必须有这个前提!

✱ **Coordinatewise Gaussian does *not* imply multivariate Gaussian.** Having each coordinate Gaussian is not enough for joint Gaussianity; the dependence structure still matters.

✱ **Example 33** (Gaussian marginals, non-Gaussian joint). Let  $Z \sim N(0, 1)$  and let  $S$  be independent of  $Z$  with  $\mathbb{P}(S = 1) = \mathbb{P}(S = -1) = \frac{1}{2}$ . Define

$$X = Z, \quad Y = SZ.$$

Then  $X \sim N(0, 1)$  and  $Y \sim N(0, 1)$  (a random sign does not change a  $N(0, 1)$  law). Moreover,

$$\text{Cov}(X, Y) = \mathbb{E}[XY] = \mathbb{E}[SZ^2] = \mathbb{E}[S]\mathbb{E}[Z^2] = 0,$$

so  $X$  and  $Y$  are uncorrelated. However, they are **not independent**: indeed  $Y^2 = X^2$ , hence  $\mathbb{P}(Y^2 = X^2) = 1$ . Consequently  $(X, Y)$  is not bivariate normal, because within the Gaussian family zero covariance forces independence.

✱ **Theorem (projection characterization of the multivariate normal).** A random vector  $X \in \mathbb{R}^p$  is multivariate normal  $N_p(\mu, \Sigma)$  if and only if for every nonzero  $c \in \mathbb{R}^p$  the scalar  $c^T X$  is (univariate) normal.

*Remark.* The full proof is beyond our scope, but you can view this as a precise version of the “all linear combinations are Gaussian” heuristic.

**Example 34.** If  $X \sim N_p(\mu, \Sigma)$  and  $c \in \mathbb{R}^p$ , then 理解:  $c^T X \propto X$  在  $c$  上的投影长度, 该长度应服从 normal,  $\forall c$

$$c^T X \sim N(c^T \mu, c^T \Sigma c).$$

Thus any 1D projection of a multivariate normal is normal with variance given by the quadratic form  $c^T \Sigma c$ .

**Example 35.** Revisit Example 33. For  $c = (1, 1)^T$ ,

$$c^T X = X_1 + X_2 = (1 + S)Z = \begin{cases} 2Z, & \text{with prob. } \frac{1}{2}, \\ 0, & \text{with prob. } \frac{1}{2}, \end{cases}$$

a mixture of a point mass at 0 and  $N(0, 4)$ ; this is not Gaussian. Hence  $X$  is not multivariate normal even though both coordinates are  $N(0, 1)$  and uncorrelated.

# 19 Lecture 20

## Expected norm and basic concentration for Gaussian vectors

We next quantify the size of a Gaussian vector. Let  $Z \sim N_p(0, I_p)$  and, more generally, let  $X \sim N_p(\mu, \Sigma)$  with  $\Sigma \succeq 0$ .

**Reminder (chi-square).** If  $X_1, \dots, X_p$  are independent  $N(0, 1)$ , then  $\sum_{i=1}^p X_i^2 \sim \chi_p^2$  (the chi-square distribution with  $p$  degrees of freedom). It satisfies

$$\mathbb{E}[\chi_p^2] = p, \quad \text{Var}(\chi_p^2) = 2p. \quad (= p \cdot \text{Var}(X_i^2) = p \cdot (\underbrace{\mathbb{E}[X_i^4]}_{=3} - \underbrace{\mathbb{E}[X_i^2]^2}_{=1}))$$

In particular, for  $Z \sim N_p(0, I_p)$ ,  $\|Z\|^2 = \sum_{i=1}^p Z_i^2 \sim \chi_p^2$ .

### Expected squared norm.

$$\mathbb{E}[\|Z\|^2] = p, \quad \text{Var}(\|Z\|^2) = 2p.$$

More generally, for  $X \sim N_p(\mu, \Sigma)$ ,

$$\mathbb{E}[\|X\|^2] = \|\mu\|^2 + \text{Tr}(\Sigma), \quad \mathbb{E}[\|X - \mu\|^2] = \text{Tr}(\Sigma).$$

Indeed, since  $\|X\|^2 = \text{Tr}(XX^T)$ , 另一种证法:  $\mathbb{E}[\|X\|^2] = \mathbb{E}[X^T X] = \mathbb{E}[(X - \mathbb{E}X + \mathbb{E}X)^T (X - \mathbb{E}X + \mathbb{E}X)] = \mathbb{E}[(X - \mathbb{E}X)^T (X - \mathbb{E}X)] + \mathbb{E}\|X\|^2$   
 $= \text{Tr}((X - \mathbb{E}X)^T (X - \mathbb{E}X)) = \text{Tr}((X - \mathbb{E}X)(X - \mathbb{E}X)^T)$   
 $\mathbb{E}\|X\|^2 = \text{Tr}(\mathbb{E}[XX^T]) = \text{Tr}(\text{Cov}(X) + \mathbb{E}[X]\mathbb{E}[X]^T) = \text{Tr}(\Sigma) + \|\mu\|^2.$

**Elementary concentration via Markov.** Applying Markov's inequality to the non-negative variable  $\|\cdot\|^2$ , for any  $r > 0$ ,

$$\mathbb{P}(\|Z\| \geq r) \leq \frac{\mathbb{E}\|Z\|^2}{r^2} = \frac{p}{r^2}, \quad \mathbb{P}(\|X - \mu\| \geq r) \leq \frac{\mathbb{E}\|X - \mu\|^2}{r^2} = \frac{\text{Tr}(\Sigma)}{r^2} \quad (\text{若 } r = \sqrt{p \cdot \text{Tr}(\Sigma)}, \text{ 则 } \|X - \mu\|_2 < r \text{ w.p. } \geq 1 - \frac{1}{p} = 0.99)$$

These are very rough but already show that the norm concentrates around  $\sqrt{p}$  (isotropic case) or around the scale  $\sqrt{\text{Tr}(\Sigma)}$  in general.

## Covariance ellipsoid and principal directions

通过 Markov ineq. 可以得到一个圆形的 bound, 但我们希望得到一个更 strict 的椭圆形的 bound.

To visualize the geometry of  $N_p(\mu, \Sigma)$ , consider the **covariance (Mahalanobis) ellipsoid** of radius  $r > 0$ :

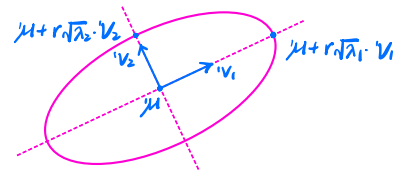
$$\mathcal{E}_r(\mu, \Sigma) := \{x \in \mathbb{R}^p : (x - \mu)^T \Sigma^{-1} (x - \mu) \leq r^2\}.$$

Diagonalize  $\Sigma = Q\Lambda Q^T$  with  $Q = [v_1 \dots v_p]$  orthogonal and  $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_p)$ ,  $\lambda_i > 0$ . Then

$$(x - \mu)^T \Sigma^{-1} (x - \mu) = \sum_{i=1}^p \frac{(v_i^T (x - \mu))^2}{\lambda_i},$$

so  $\mathcal{E}_r(\mu, \Sigma)$  is the image of the Euclidean ball  $\{u : \|u\| \leq r\}$  under  $u \mapsto \mu + Q\Lambda^{1/2}u$ . Concretely:

$$\begin{aligned} (x - \mu)^T \Sigma^{-1} (x - \mu) &\leq r^2 \\ \Rightarrow (x - \mu)^T (Q^{-1})^T \Lambda^{-1/2} \Lambda^{-1/2} Q^{-1} (x - \mu) &\leq r^2 \\ \Rightarrow \|\Lambda^{-1/2} Q^{-1} (x - \mu)\|_2 &\leq r \\ \text{令 } u &:= \Lambda^{-1/2} Q^{-1} (x - \mu) \\ \Rightarrow x &= \mu + \Lambda^{1/2} Q u \end{aligned}$$



- The principal axes of the ellipsoid are the eigenvectors  $v_i$ .
- The semi-axis length along  $v_i$  equals  $r\sqrt{\lambda_i}$ . 令  $u = r \cdot v_i$ , 则 length 为  $\| \Sigma^{1/2} \cdot u \|_2 = r \cdot \sqrt{\lambda_i}$
- Large  $\lambda_i$  indicate directions of high spread; small  $\lambda_i$  indicate directions of concentration.

**Probability content of ellipsoids.** If  $X \sim N_p(\mu, \Sigma)$ , then the squared Mahalanobis distance  $(X - \mu)^T \Sigma^{-1} (X - \mu)$  has the  $\chi_p^2$  distribution. Therefore

$$= (\Sigma^{-1/2} (X - \mu))^T (\Sigma^{-1/2} (X - \mu))$$

$$\mathbb{P}(X \in \mathcal{E}_r(\mu, \Sigma)) = \mathbb{P}(\chi_p^2 \leq r^2).$$

$$\begin{aligned} X - \mu &\sim N_d(0, \Sigma) \\ \Rightarrow \Sigma^{-1/2} (X - \mu) &\sim N_d(0, \Sigma^{-1/2} \Sigma \Sigma^{-1/2}) \\ &\sim N_d(0, I_d) \end{aligned}$$

Choosing  $r^2$  as a  $\chi_p^2$  quantile produces the familiar concentric ellipsoidal confidence/contour regions of the Gaussian density.

**Example 36** (Directional variances and axes). For any unit vector  $v$ , the variance of the projection is

$$\text{Var}(v^T X) = v^T \Sigma v = \sum_{i=1}^p \lambda_i (v^T v_i)^2.$$

$$\begin{aligned} \text{若 } v \text{ 也为某 eigenvector } v_i: \\ v_i^T \Sigma v_i &= v_i^T v_i \lambda_i v_i^T v_i = \lambda_i \|v_i\|_2^4 = \lambda_i \\ v_i^T X &\sim N(0, \lambda_i) \end{aligned}$$

The maximal and minimal directional variances are  $\lambda_{\max}$  and  $\lambda_{\min}$ , attained at the corresponding eigenvectors. Thus the longest axis of  $\mathcal{E}_r(\mu, \Sigma)$  points in  $v_{\max}$  with length  $r\sqrt{\lambda_{\max}}$ , while the shortest points in  $v_{\min}$  with length  $r\sqrt{\lambda_{\min}}$ .

## Lectures 21 and 22

### Application: Gaussian linear regression

We briefly show how the multivariate normal machinery applies to the standard multiple linear regression model. In this section we treat the design matrix  $X$  as fixed (non-random).

#### Model and notation

We observe  $n$  response values  $y_1, \dots, y_n$  and covariate vectors  $x_i \in \mathbf{R}^p$  (often with the first component equal to 1 for an intercept). Collect these into

$$Y = \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} \in \mathbf{R}^n, \quad X = \begin{pmatrix} 1 & x_{11} & \dots & x_{1p} \\ 1 & x_{21} & \dots & x_{2p} \\ \vdots & \vdots & & \vdots \\ 1 & x_{n1} & \dots & x_{np} \end{pmatrix} \in \mathbf{R}^{n \times (p+1)}.$$

(For simplicity we will just write  $p$  for the number of regression coefficients, so  $X$  is  $n \times p$  and  $\beta \in \mathbf{R}^p$ .)

The *Gaussian linear regression model* assumes

$$Y = X\beta + \varepsilon,$$

where  $\beta \in \mathbf{R}^p$  is an unknown parameter vector and the error vector  $\varepsilon \in \mathbf{R}^n$  satisfies

$$\mathbb{E}[\varepsilon] = \mathbf{0}, \quad \text{Cov}(\varepsilon) = \sigma^2 I_n.$$

Equivalently,

$$Y \sim N_n(X\beta, \sigma^2 I_n).$$

### Mean and covariance of $Y$

From the model,

$$\mathbb{E}[Y] = \mathbb{E}[X\beta + \varepsilon] = X\beta + \mathbb{E}[\varepsilon] = X\beta,$$

and, using basic covariance rules and the fact that  $X\beta$  is deterministic,

$$\text{Cov}(Y) = \text{Cov}(X\beta + \varepsilon) = \text{Cov}(\varepsilon) = \sigma^2 I_n.$$

### Least squares estimator

The (ordinary) least squares estimator of  $\beta$  is the vector  $\hat{\beta}$  that minimizes the residual sum of squares  $\|Y - Xb\|^2$  over  $b \in \mathbf{R}^p$ . Linear algebra shows that if  $X^\top X$  is invertible, the minimizer is unique and given by

$$\hat{\beta} = (X^\top X)^{-1} X^\top Y.$$

### Expectation, covariance, and distribution of $\hat{\beta}$

**Expectation (unbiasedness).** Using  $\mathbb{E}[Y] = X\beta$ ,

$$\mathbb{E}[\hat{\beta}] = \mathbb{E}[(X^\top X)^{-1} X^\top Y] = (X^\top X)^{-1} X^\top \mathbb{E}[Y] = (X^\top X)^{-1} X^\top X\beta = \beta.$$

Thus  $\hat{\beta}$  is an unbiased estimator of  $\beta$ .

**Covariance matrix.** Because  $\hat{\beta}$  is a linear function of  $Y$ ,

$$\text{Cov}(\hat{\beta}) = \text{Cov}((X^\top X)^{-1} X^\top Y) = (X^\top X)^{-1} X^\top \text{Cov}(Y) X (X^\top X)^{-1}.$$

Substituting  $\text{Cov}(Y) = \sigma^2 I_n$  gives

$$\text{Cov}(\hat{\beta}) = \sigma^2 (X^\top X)^{-1} X^\top X (X^\top X)^{-1} = \sigma^2 (X^\top X)^{-1}.$$

**Distribution.** We know that  $Y \sim N_n(X\beta, \sigma^2 I_n)$ . The estimator  $\hat{\beta}$  is a linear transformation of  $Y$ :  $\hat{\beta} = BY$  with  $B = (X^\top X)^{-1} X^\top$ . By stability of multivariate normals under linear maps,

$$\hat{\beta} \sim N_p(BX\beta, B(\sigma^2 I_n)B^\top) = N_p(\beta, \sigma^2 (X^\top X)^{-1}). \quad \Rightarrow \quad \hat{\beta}_j \sim \mathcal{N}(\beta_j, (\sigma^2 (X^\top X)^{-1})_{jj})$$

$= e_j^\top \hat{\beta} \quad \quad \quad = e_j^\top \Sigma_{\hat{\beta}} e_j$

In short,

$$\hat{\beta} \sim N_p(\beta, \sigma^2 (X^\top X)^{-1}).$$

## Confidence ellipsoid for $\beta$

This is a direct special case of our covariance-ellipsoid discussion. Set

$$\Sigma_{\hat{\beta}} = \sigma^2 (X^\top X)^{-1}.$$

Then  $\hat{\beta} \sim N_p(\beta, \Sigma_{\hat{\beta}})$ . For a general Gaussian vector  $W \sim N_p(0, \Sigma)$  we know that

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$$W^\top \Sigma^{-1} W \sim \chi_p^2.$$

Applying this with  $W = \hat{\beta} - \beta$  and  $\Sigma = \Sigma_{\hat{\beta}}$  gives

$$(\hat{\beta} - \beta)^\top \Sigma_{\hat{\beta}}^{-1} (\hat{\beta} - \beta) \sim \chi_p^2.$$

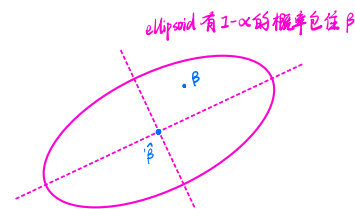
Since  $\Sigma_{\hat{\beta}}^{-1} = \frac{1}{\sigma^2} X^\top X$ , we obtain



$$\frac{1}{\sigma^2} (\hat{\beta} - \beta)^\top X^\top X (\hat{\beta} - \beta) \sim \chi_p^2.$$

Let  $r^2 = \chi_{p, 1-\alpha}^2$  be the  $(1 - \alpha)$ -quantile of the  $\chi_p^2$  distribution. Then

$$\mathbb{P}\left(\frac{1}{\sigma^2} (\hat{\beta} - \beta)^\top X^\top X (\hat{\beta} - \beta) \leq r^2\right) = 1 - \alpha.$$



Equivalently,

$$\mathbb{P}(\beta \in \mathcal{C}_{1-\alpha}) = 1 - \alpha,$$

where the  $(1 - \alpha)$ -confidence ellipsoid for  $\beta$  is

$$\mathcal{C}_{1-\alpha} = \left\{ b \in \mathbf{R}^p : (b - \hat{\beta})^\top X^\top X (b - \hat{\beta}) \leq \underbrace{\sigma^2 \chi_{p, 1-\alpha}^2} \right\}.$$

This is exactly the covariance ellipsoid for a  $N_p(\beta, \Sigma_{\hat{\beta}})$  vector, centered at  $\hat{\beta}$  with shape matrix  $\Sigma_{\hat{\beta}}$ . Its principal axes are the eigenvectors of  $X^\top X$ , and the axis lengths are proportional to the square roots of the eigenvalues of  $(X^\top X)^{-1}$ .

## Marginals, conditionals, and the linear-Gaussian regression model

Let  $(X, Y)$  be a random vector taking values in  $\mathbf{R}^p \times \mathbf{R}^q$  with joint density  $f_{X,Y}(x, y)$ .

**Reminder: Marginals and conditionals (vector case).**

$$f_X(x) = \int_{\mathbf{R}^q} f_{X,Y}(x, y) dy, \quad f_Y(y) = \int_{\mathbf{R}^p} f_{X,Y}(x, y) dx,$$

and, whenever  $f_X(x) > 0$ ,

$$f_{Y|X=x}(y) = \frac{f_{X,Y}(x, y)}{f_X(x)}.$$

For any function  $g : \mathbf{R}^q \rightarrow \mathbf{R}$  such that the integrals exist,

$$\mathbb{E}[g(Y) \mid X = x] = \int_{\mathbf{R}^q} g(y) f_{Y|X=x}(y) dy.$$

**Block Gaussian identities (conditioning and regression).** Assume

$$\begin{pmatrix} X \\ Y \end{pmatrix} \sim N_{p+q} \left( \begin{pmatrix} \mu_X \\ \mu_Y \end{pmatrix}, \begin{pmatrix} \Sigma_{XX} & \Sigma_{XY} \\ \Sigma_{YX} & \Sigma_{YY} \end{pmatrix} \right),$$

where  $\Sigma_{XX}$  is invertible. Then the conditional distribution of  $Y$  given  $X = x$  is Gaussian with

$$\begin{aligned} \mathbb{E}[Y \mid X = x] &= \mu_Y + \Sigma_{YX} \Sigma_{XX}^{-1} (x - \mu_X), \\ \text{Cov}(Y \mid X = x) &= \Sigma_{YY} - \Sigma_{YX} \Sigma_{XX}^{-1} \Sigma_{XY}. \end{aligned}$$

Equivalently, there are  $A \in \mathbf{R}^{q \times p}$  and  $b \in \mathbf{R}^q$  such that

$$Y \mid X = x \sim N_q(Ax + b, \Sigma_{Y|X}), \quad \begin{cases} A = \Sigma_{YX} \Sigma_{XX}^{-1}, \\ b = \mu_Y - A\mu_X, \\ \Sigma_{Y|X} = \Sigma_{YY} - \Sigma_{YX} \Sigma_{XX}^{-1} \Sigma_{XY}. \end{cases}$$

So, in the Gaussian world, the conditional mean is *automatically linear* in  $x$  and the conditional covariance does not depend on  $x$ .

We now make the linear structure even more explicit by writing down a concrete linear transformation that produces the regression residuals.

**Residuals as a linear transform of  $(X, Y)$ .** Assume

$$\begin{pmatrix} X \\ Y \end{pmatrix} \sim N_{p+q} \left( \begin{pmatrix} \mu_X \\ \mu_Y \end{pmatrix}, \begin{pmatrix} \Sigma_{XX} & \Sigma_{XY} \\ \Sigma_{YX} & \Sigma_{YY} \end{pmatrix} \right), \quad \Sigma_{XX} \text{ invertible},$$

and set

$$A = \Sigma_{YX} \Sigma_{XX}^{-1}, \quad \varepsilon = Y - \mu_Y - A(X - \mu_X).$$

Write the centered vector as

$$Z = \begin{pmatrix} X - \mu_X \\ Y - \mu_Y \end{pmatrix}, \quad \text{Cov}(Z) = \begin{pmatrix} \Sigma_{XX} & \Sigma_{XY} \\ \Sigma_{YX} & \Sigma_{YY} \end{pmatrix}.$$

Define the linear map

$$T = \begin{pmatrix} I_p & 0 \\ -A & I_q \end{pmatrix}, \quad \begin{pmatrix} X - \mu_X \\ \varepsilon \end{pmatrix} = T \begin{pmatrix} X - \mu_X \\ Y - \mu_Y \end{pmatrix}.$$

Since  $T$  is invertible and  $Z$  is Gaussian, the transformed vector  $(X - \mu_X, \varepsilon)$  is also jointly Gaussian.

*Means.* From the definition,

$$\mathbb{E}[\varepsilon] = \mathbb{E}[Y - \mu_Y - A(X - \mu_X)] = \mu_Y - \mu_Y - A(\mu_X - \mu_X) = 0.$$

*Covariances via the transform.* The covariance matrix of  $(X - \mu_X, \varepsilon)$  is

$$\text{Cov}\begin{pmatrix} X - \mu_X \\ \varepsilon \end{pmatrix} = T \begin{pmatrix} \Sigma_{XX} & \Sigma_{XY} \\ \Sigma_{YX} & \Sigma_{YY} \end{pmatrix} T^\top.$$

Multiplying out the blocks,

$$T \begin{pmatrix} \Sigma_{XX} & \Sigma_{XY} \\ \Sigma_{YX} & \Sigma_{YY} \end{pmatrix} = \begin{pmatrix} \Sigma_{XX} & \Sigma_{XY} \\ \Sigma_{YX} - A\Sigma_{XX} & \Sigma_{YY} - A\Sigma_{XY} \end{pmatrix},$$

and then

$$\text{Cov}\begin{pmatrix} X - \mu_X \\ \varepsilon \end{pmatrix} = \begin{pmatrix} \Sigma_{XX} & \Sigma_{XY} - \Sigma_{XX}A^\top \\ \Sigma_{YX} - A\Sigma_{XX} & \Sigma_{YY} - A\Sigma_{XY} - \Sigma_{YX}A^\top + A\Sigma_{XX}A^\top \end{pmatrix}.$$

With  $A = \Sigma_{YX}\Sigma_{XX}^{-1}$  and  $\Sigma_{XY} = \Sigma_{YX}^\top$  we get

$$\Sigma_{YX} - A\Sigma_{XX} = \Sigma_{YX} - \Sigma_{YX}\Sigma_{XX}^{-1}\Sigma_{XX} = 0,$$

and similarly  $\Sigma_{XY} - \Sigma_{XX}A^\top = 0$ . Thus the covariance is block-diagonal:

$$\text{Cov}\begin{pmatrix} X - \mu_X \\ \varepsilon \end{pmatrix} = \begin{pmatrix} \Sigma_{XX} & 0 \\ 0 & \Sigma_{YY} - \Sigma_{YX}\Sigma_{XX}^{-1}\Sigma_{XY} \end{pmatrix}.$$

In particular

$$\text{Cov}(\varepsilon) = \Sigma_{YY} - \Sigma_{YX}\Sigma_{XX}^{-1}\Sigma_{XY},$$

and  $\text{Cov}(\varepsilon, X) = 0$ . Because  $(X - \mu_X, \varepsilon)$  is jointly Gaussian and uncorrelated,  $X$  and  $\varepsilon$  are in fact independent.

*Conditioning.* From the definition of  $\varepsilon$ ,

$$Y = \mu_Y + A(X - \mu_X) + \varepsilon, \quad \varepsilon \perp\!\!\!\perp X, \quad \varepsilon \sim N_q(0, \Sigma_{Y|X}).$$

Conditioning on  $X = x$  freezes the deterministic part and leaves the independent noise:

$$Y \mid X = x \sim N_q(\mu_Y + A(x - \mu_X), \Sigma_{Y|X}),$$

so

$$\boxed{\mathbb{E}[Y \mid X = x] = \mu_Y + \Sigma_{YX}\Sigma_{XX}^{-1}(x - \mu_X), \quad \text{Cov}(Y \mid X = x) = \Sigma_{YY} - \Sigma_{YX}\Sigma_{XX}^{-1}\Sigma_{XY}.}$$

Thus, believing that  $(X, Y)$  is jointly Gaussian *forces* a linear regression structure plus independent Gaussian residuals.

**Example 37** (Population linear regression as a conditional mean). Let  $X \in \mathbf{R}^p$  be a vector of covariates and  $Y \in \mathbf{R}$  a scalar response. We are in the block setup above with  $q = 1$ , so  $\Sigma_{YY}$  is a scalar and  $\Sigma_{XY}$  is a column vector. From the formulas,

$$\mathbb{E}[Y \mid X = x] = \mu_Y + \Sigma_{YX} \Sigma_{XX}^{-1} (x - \mu_X), \quad \text{Var}(Y \mid X = x) = \Sigma_{YY} - \Sigma_{YX} \Sigma_{XX}^{-1} \Sigma_{XY}.$$

Define

$$\beta := \Sigma_{XX}^{-1} \Sigma_{XY} \in \mathbf{R}^p, \quad \beta_0 := \mu_Y - \beta^\top \mu_X \in \mathbf{R}, \quad \sigma^2 := \Sigma_{YY} - \Sigma_{YX} \Sigma_{XX}^{-1} \Sigma_{XY}.$$

Then

$$\mathbb{E}[Y \mid X = x] = \beta_0 + \beta^\top x,$$

and if we set  $\varepsilon := Y - \beta_0 - \beta^\top X$ , we can write

$$Y_{|X=x} = \beta_0 + \beta^\top x + \varepsilon, \quad \varepsilon \sim N(0, \sigma^2), \quad \varepsilon \perp X.$$

So, under the joint Gaussian assumption, the *population* regression function is exactly linear in  $x$ , and the deviations from this line are independent Gaussian errors.

## Introduction to Bayesian linear regression: Gaussian, Laplace and diffuse priors

We now return to the Gaussian linear regression model, but take a Bayesian viewpoint. The idea is that the regression coefficients  $\beta$  are no longer fixed unknown numbers, but a random vector with some prior distribution. The Gaussian likelihood together with different choices of prior will naturally lead us to familiar penalized estimators such as ridge and lasso.

**Model and likelihood (no intercept for simplicity).** We observe responses  $y_1, \dots, y_n$  and covariate vectors  $x_i \in \mathbf{R}^p$ , collected into

$$Y = \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} \in \mathbf{R}^n, \quad X = \begin{pmatrix} x_{11} & \dots & x_{1p} \\ \vdots & & \vdots \\ x_{n1} & \dots & x_{np} \end{pmatrix} \in \mathbf{R}^{n \times p}.$$

(An intercept can always be included by adding a column of 1's to  $X$ ; we drop it in the notation to keep the formulas lighter.)

The Gaussian linear regression model is

$$Y = X\beta + \varepsilon, \quad \varepsilon \sim N_n(0, \sigma^2 I_n),$$

where  $\beta \in \mathbf{R}^p$  and  $\sigma > 0$  are unknown. Conditional on  $(\beta, \sigma)$ ,

$$Y \mid (\beta, \sigma) \sim N_n(X\beta, \sigma^2 I_n),$$



so the likelihood (as a function of  $\beta, \sigma$ ) is

$$L(\beta, \sigma; Y) = \frac{1}{(\sqrt{2\pi}\sigma)^n} \exp\left(-\frac{1}{2\sigma^2}\|Y - X\beta\|^2\right),$$

where

$$S(\beta) := \|Y - X\beta\|^2 = (Y - X\beta)^\top (Y - X\beta)$$

is the usual residual sum of squares. In what follows we will mostly treat  $\sigma > 0$  as known, to focus on how the choice of prior on  $\beta$  leads to different estimators.

### Gaussian priors on $\beta$ and ridge regression

A very natural Bayesian choice is a Gaussian prior on  $\beta$ :

$$\beta \sim N_p(0, \tau^2 I_p),$$

with  $\tau > 0$  a prior scale. This encodes the informal belief that the coefficients are centered near 0 and of size on the order of  $\tau$ . The prior density is

$$\pi(\beta) = \frac{1}{(\sqrt{2\pi}\tau)^p} \exp\left(-\frac{1}{2\tau^2}\|\beta\|^2\right) \propto \exp\left(-\frac{1}{2\tau^2}\|\beta\|^2\right).$$

By Bayes' rule (treating  $\sigma$  as fixed),

$$\pi(\beta | Y) \propto L(\beta, \sigma; Y) \pi(\beta) \propto \exp\left(-\frac{1}{2\sigma^2}S(\beta)\right) \exp\left(-\frac{1}{2\tau^2}\|\beta\|^2\right).$$

Hence

$$\log \pi(\beta | Y) = \text{const} - \frac{1}{2\sigma^2}S(\beta) - \frac{1}{2\tau^2}\|\beta\|^2,$$

so up to an additive constant the negative log-posterior is

$$\frac{1}{2\sigma^2}S(\beta) + \frac{1}{2\tau^2}\|\beta\|^2.$$

The *maximum a posteriori* (MAP) estimator chooses the  $\beta$  with the largest posterior density, or equivalently the smallest negative log-posterior:

$$\hat{\beta}_{\text{MAP}} = \arg \min_{\beta \in \mathbf{R}^p} \left\{ \frac{1}{2\sigma^2}S(\beta) + \frac{1}{2\tau^2}\|\beta\|^2 \right\}.$$

Multiplying the objective by  $2\sigma^2$  does not change the minimizer, so

$$\hat{\beta}_{\text{MAP}} = \arg \min_{\beta \in \mathbf{R}^p} \left\{ S(\beta) + \lambda \|\beta\|^2 \right\}, \quad \lambda = \frac{\sigma^2}{\tau^2} > 0.$$

This is exactly the *ridge regression* estimator with tuning parameter  $\lambda$ . So:

|                                                                                  |
|----------------------------------------------------------------------------------|
| Gaussian prior on $\beta$ $\iff$ ridge ( $\ell_2$ -penalized) regression as MAP. |
|----------------------------------------------------------------------------------|

NOTE: The materials presented below (until the end of the lecture) are supplementary and will not be covered in the lecture, as these examples have not yet been studied in the statistics classes. Nevertheless, it is recommended that you familiarize yourself with them.

### Laplace priors on $\beta$ and lasso

To encourage *sparsity* (many coefficients exactly zero), a popular choice is an independent Laplace prior on the coordinates of  $\beta$ . For a single coordinate  $\beta_j$ ,

$$\pi(\beta_j) = \frac{\lambda}{2} e^{-\lambda|\beta_j|}, \quad j = 1, \dots, p,$$

with  $\lambda > 0$  a prior scale. Assuming independence across  $j$ ,

$$\pi(\beta) = \prod_{j=1}^p \frac{\lambda}{2} e^{-\lambda|\beta_j|} \propto \exp(-\lambda\|\beta\|_1), \quad \|\beta\|_1 = \sum_{j=1}^p |\beta_j|.$$

**Posterior and MAP.** Combining this with the Gaussian likelihood,

$$\pi(\beta \mid Y) \propto \exp\left(-\frac{1}{2\sigma^2}S(\beta)\right) \exp(-\lambda\|\beta\|_1),$$

so

$$-\log \pi(\beta \mid Y) = \text{const} + \frac{1}{2\sigma^2}S(\beta) + \lambda\|\beta\|_1.$$

The MAP estimator therefore minimizes

$$\hat{\beta}_{\text{MAP}} = \arg \min_{\beta \in \mathbf{R}^p} \left\{ \frac{1}{2\sigma^2}S(\beta) + \lambda\|\beta\|_1 \right\}.$$

Multiplying by  $2\sigma^2$ ,

$$\hat{\beta}_{\text{MAP}} = \arg \min_{\beta \in \mathbf{R}^p} \{S(\beta) + \lambda_{\text{lasso}}\|\beta\|_1\}, \quad \lambda_{\text{lasso}} = 2\sigma^2\lambda.$$

This is precisely the *lasso* objective: least-squares loss plus an  $\ell_1$  penalty. The exact numerical value of  $\lambda_{\text{lasso}}$  is not as important as the shape: an  $\ell_1$  penalty arises directly from a Laplace prior.

|                                                                                 |
|---------------------------------------------------------------------------------|
| Laplace prior on $\beta$ $\iff$ lasso ( $\ell_1$ -penalized) regression as MAP. |
|---------------------------------------------------------------------------------|

Unlike the Gaussian prior case, the posterior  $\beta \mid Y$  is no longer Gaussian, and there is no simple closed form for its full distribution. But the MAP estimator still has a clean optimization interpretation.

### Example: Diffuse priors on $(\beta, \sigma)$

For comparison, we recall the “almost flat” prior treated earlier, which leads back to (approximately) least-squares without any penalty.

We again work with

$$Y = X\beta + \varepsilon, \quad \varepsilon \sim N_n(0, \sigma^2 I_n),$$

and now place a weak prior on both  $\beta$  and  $\sigma$ . Fix a large constant  $C > 0$  and take

$$\beta_j \sim \text{Unif}(-C, C) \quad (j = 1, \dots, p), \quad \log \sigma \sim \text{Unif}(-C, C),$$

independent. The corresponding joint prior density is

$$\pi_C(\beta, \sigma) \propto \frac{1}{\sigma} \mathbf{1}\{-C < \beta_j < C \ \forall j, \ e^{-C} < \sigma < e^C\}.$$

**Posterior for  $(\beta, \sigma)$ .** Bayes' rule gives, up to a normalizing constant,

$$\pi_C(\beta, \sigma \mid Y) \propto L(\beta, \sigma; Y) \pi_C(\beta, \sigma) \propto \sigma^{-n-1} \exp\left(-\frac{S(\beta)}{2\sigma^2}\right) \mathbf{1}\{-C < \beta_j < C, \ e^{-C} < \sigma < e^C\}.$$

**Marginal posterior for  $\beta$ .** To see how the data inform  $\beta$ , integrate out  $\sigma$ :

$$\pi_C(\beta \mid Y) \propto \mathbf{1}\{-C < \beta_j < C\} \int_{e^{-C}}^{e^C} \sigma^{-n-1} \exp\left(-\frac{S(\beta)}{2\sigma^2}\right) d\sigma.$$

For large  $C$  the main contribution of the integral comes from moderate values of  $\sigma$ , so the tails outside  $(e^{-C}, e^C)$  are negligible. Thus, for large  $C$ ,

$$\int_{e^{-C}}^{e^C} \dots d\sigma \approx \int_0^\infty \sigma^{-n-1} \exp\left(-\frac{S(\beta)}{2\sigma^2}\right) d\sigma.$$

Let  $a = S(\beta)/2 > 0$  and consider

$$I(a) = \int_0^\infty \sigma^{-n-1} e^{-a/\sigma^2} d\sigma.$$

With the change of variable  $u = a/\sigma^2$  (so  $\sigma = \sqrt{a/u}$  and  $d\sigma = -\frac{1}{2}a^{1/2}u^{-3/2}du$ ), we get

$$\begin{aligned} I(a) &= \int_\infty^0 \left(\frac{a}{u}\right)^{-(n+1)/2} e^{-u} \left(-\frac{1}{2}a^{1/2}u^{-3/2}\right) du \\ &= \frac{1}{2}a^{-n/2} \int_0^\infty u^{n/2-1} e^{-u} du \propto a^{-n/2}, \end{aligned}$$

and the remaining Gamma integral does not depend on  $a$ . Therefore, for large  $C$ ,

$$\pi_C(\beta \mid Y) \propto \mathbf{1}\{-C < \beta_j < C\} S(\beta)^{-n/2}.$$

As  $C$  is taken very large, the posterior mass concentrates well inside the box  $\{-C < \beta_j < C\}$ , so on the scale where the posterior is non-negligible we may (heuristically) drop this indicator and write

$$\pi(\beta \mid Y) \propto S(\beta)^{-n/2} = \|Y - X\beta\|_2^{-n}.$$

**Connection to least squares and concentration.** Recall that the least-squares estimator

$$\hat{\beta} = (X^T X)^{-1} X^T Y$$

is the unique minimizer of  $S(\beta)$  when  $X^T X$  is invertible, and that

$$S(\beta) = S(\hat{\beta}) + (\beta - \hat{\beta})^T X^T X (\beta - \hat{\beta})$$

(by expanding squares). Plugging this into the posterior shape gives, for large  $C$ ,

$$\pi(\beta | Y) \propto \left( S(\hat{\beta}) + (\beta - \hat{\beta})^T X^T X (\beta - \hat{\beta}) \right)^{-n/2}.$$

Thus:

- The posterior mode (point of highest density) is at  $\beta = \hat{\beta}$ ; the Bayesian MAP estimate coincides with the least-squares estimator.
- If  $S(\beta) = (1 + \delta)S(\hat{\beta})$  with  $\delta > 0$ , then

$$\frac{\pi(\beta | Y)}{\pi(\hat{\beta} | Y)} \approx \left( \frac{S(\hat{\beta})}{S(\beta)} \right)^{n/2} = \left( \frac{1}{1 + \delta} \right)^{n/2},$$

which is tiny when  $n$  is large unless  $\delta$  is extremely small. So the posterior mass is sharply concentrated on those  $\beta$  for which  $S(\beta)$  is very close to its minimum  $S(\hat{\beta})$ .

In summary: Gaussian priors lead to ridge, Laplace priors lead to lasso, and a very flat prior leads back to least-squares. All of these familiar regression procedures can be viewed as different MAP estimators in the same Gaussian linear model.

## Lecture 23

### Order statistics

In many problems we do not care about the whole sample, but only about the *extremes* or the *ranked* values: the minimum, maximum, median, etc. Let  $X_1, \dots, X_n$  be i.i.d. real-valued r.v.'s with cdf  $F$  and density  $f$ . Sort the sample:

$$X_{(1)} \leq X_{(2)} \leq \dots \leq X_{(n)}.$$

The r.v.  $X_{(k)}$  is called the *kth order statistic*. In particular,

$$X_{(1)} = \min_i X_i, \quad X_{(n)} = \max_i X_i.$$

## Distribution of the maximum and minimum

Let  $M_n := X_{(n)} = \max(X_1, \dots, X_n)$  and  $m_n := X_{(1)} = \min(X_1, \dots, X_n)$ .

**Maximum.** For any real  $t$ ,

$$\mathbb{P}\{M_n \leq t\} = \mathbb{P}\{X_1 \leq t, \dots, X_n \leq t\} = \prod_{i=1}^n \mathbb{P}\{X_i \leq t\} = F(t)^n,$$

by independence and identical distribution. So the cdf of  $M_n$  is

$$F_{M_n}(t) = F(t)^n.$$

If  $F$  has density  $f$ , then differentiating gives

$$f_{M_n}(t) = \frac{d}{dt} F(t)^n = nF(t)^{n-1}f(t).$$

**Minimum.** Similarly,

$$\mathbb{P}\{m_n > t\} = \mathbb{P}\{X_1 > t, \dots, X_n > t\} = (1 - F(t))^n,$$

so

$$F_{m_n}(t) = \mathbb{P}\{m_n \leq t\} = 1 - (1 - F(t))^n,$$

and, when  $F$  has a density  $f$ ,

$$f_{m_n}(t) = \frac{d}{dt} F_{m_n}(t) = n(1 - F(t))^{n-1}f(t).$$

**Example 38** (Extremes of a Uniform(0, 1) sample). Let  $X_1, \dots, X_n$  be i.i.d. Unif(0, 1). Then  $F(t) = t$  and  $f(t) = 1$  on  $0 < t < 1$ .

*Maximum.* For  $0 < t < 1$ ,

$$F_{M_n}(t) = t^n, \quad f_{M_n}(t) = nt^{n-1}.$$

*Intuition:*

$$\begin{array}{c} [\dots \dots \dots] \\ 0 \quad \quad \quad 1 \\ \text{\textit{\textcolor{violet}{*n+1 intervals}}} \\ \Rightarrow \begin{cases} \mathbb{E}[M_n] = \frac{n}{n+1} \\ \mathbb{E}[m_n] = \frac{1}{n+1} \end{cases} \end{array}$$

The expectation is

$$\mathbb{E}[M_n] = \int_0^1 t f_{M_n}(t) dt = \int_0^1 t \cdot nt^{n-1} dt = n \int_0^1 t^n dt = n \cdot \frac{1}{n+1} = \frac{n}{n+1}.$$

*Minimum.* For  $0 < t < 1$ ,

$$F_{m_n}(t) = 1 - (1 - t)^n, \quad f_{m_n}(t) = n(1 - t)^{n-1}.$$

The expectation is

$$\mathbb{E}[m_n] = \int_0^1 t n(1 - t)^{n-1} dt.$$

Here it is easier to use the symmetry of the uniform:

$$m_n = 1 - M'_n,$$

where  $M'_n$  is the maximum of another i.i.d.  $\text{Unif}(0, 1)$  sample. Therefore

$$\mathbb{E}[m_n] = 1 - \mathbb{E}[M_n] = 1 - \frac{n}{n+1} = \frac{1}{n+1}.$$

**Example 39** (Minimum of exponential variables (first arrival time)). Let  $X_1, \dots, X_n$  be i.i.d. exponential with rate  $\lambda > 0$ :

$$F(t) = 1 - e^{-\lambda t}, \quad t \geq 0.$$

Then, for  $t \geq 0$ ,

$$\mathbb{P}\{m_n > t\} = (1 - F(t))^n = (e^{-\lambda t})^n = e^{-n\lambda t},$$

hence

$$F_{m_n}(t) = 1 - e^{-n\lambda t}, \quad f_{m_n}(t) = n\lambda e^{-n\lambda t}.$$

So

$$m_n \sim \text{Exp}(n\lambda), \quad \mathbb{E}[m_n] = \frac{1}{n\lambda}.$$

Interpretation: suppose  $X_i$  is the waiting time until the  $i$ th independent alarm clock rings, each with  $\text{exponential}(\lambda)$  waiting time. Then  $m_n = \min_i X_i$  is the time until the *first* alarm, and it is again exponential, with rate  $n\lambda$ .

**Example 40** (Which observation is the minimum?). Assume  $X_1, \dots, X_n$  are i.i.d. with a continuous distribution (so ties occur with probability 0). Let

$$I_{\min} = \arg \min_{1 \leq i \leq n} X_i$$

be the index of the smallest observation. By symmetry, each of the  $n$  positions is equally likely to contain the minimum:

$$\mathbb{P}\{I_{\min} = j\} = \frac{1}{n}, \quad j = 1, \dots, n.$$

Any strict ordering of  $(X_1, \dots, X_n)$  has the same probability (i.i.d. and continuous). In  $1/n$  of these orderings, the  $j$ th coordinate is the smallest. The same argument shows that the index of the maximum is also uniform on  $\{1, \dots, n\}$ . This kind of symmetry fact is useful in quick probability questions: for example, the chance that the minimum happens to be one of the last three observations is simply  $3/n$ .

## Distribution of the $k$ th order statistic

We now state the general formula for  $X_{(k)}$ .

**CDF of  $X_{(k)}$ .** Fix  $k \in \{1, \dots, n\}$ . For any real  $t$ ,

$$\{X_{(k)} \leq t\} \iff \text{at least } k \text{ of the } X_i \text{'s are } \leq t.$$

Each  $X_i$  is  $\leq t$  with probability  $F(t)$  and  $> t$  with probability  $1 - F(t)$ , independently. Thus the cdf of  $X_{(k)}$  is

$$F_{X_{(k)}}(t) = \mathbb{P}\{X_{(k)} \leq t\} = \sum_{j=k}^n \binom{n}{j} F(t)^j (1 - F(t))^{n-j}.$$

**PDF of  $X_{(k)}$  (continuous case).** Assume  $F$  has density  $f$  so that  $F'(t) = f(t)$ . Differentiate term by term:

$$\begin{aligned} f_{X_{(k)}}(t) &= \frac{d}{dt} F_{X_{(k)}}(t) \\ &= \sum_{j=k}^n \binom{n}{j} \frac{d}{dt} (F(t)^j (1 - F(t))^{n-j}) \\ &= \sum_{j=k}^n \binom{n}{j} \left[ j F(t)^{j-1} (1 - F(t))^{n-j} - (n-j) F(t)^j (1 - F(t))^{n-j-1} \right] f(t). \end{aligned}$$

Using the binomial identities  $j \binom{n}{j} = n \binom{n-1}{j-1}$  and  $(n-j) \binom{n}{j} = n \binom{n-1}{j}$ , the sum simplifies (neighboring terms cancel) to a single term:

$$f_{X_{(k)}}(t) = n \binom{n-1}{k-1} F(t)^{k-1} (1 - F(t))^{n-k} f(t) = \frac{n!}{(k-1)!(n-k)!} F(t)^{k-1} (1 - F(t))^{n-k} f(t).$$

So

$n$  个 samples 中挑  $(k-1)$  个排在  $X_{(k)}$  前,  $(n-k)$  个排在  $X_{(k)}$  后, 1 个刚好等于  $X_{(k)}$  的组合数

$$f_{X_{(k)}}(t) = \frac{n!}{(k-1)!(n-k)!} (F(t))^{k-1} (1 - F(t))^{n-k} f(t). \quad = n \binom{n-1}{k-1} F(t)^{k-1} (1 - F(t))^{n-k} f(t)$$

**Example 41** ( $k$ th order statistic for Uniform(0, 1)). Again take  $X_1, \dots, X_n$  i.i.d. Unif(0, 1), so  $F(t) = t$  and  $f(t) = 1$  on  $(0, 1)$ . Then for  $0 < t < 1$ ,

$$f_{X_{(k)}}(t) = \frac{n!}{(k-1)!(n-k)!} t^{k-1} (1-t)^{n-k}.$$

This is the density of a Beta( $k, n+1-k$ ) distribution on  $(0, 1)$ . A standard Beta integral (or  $k$  integrations by parts) shows that

$$\mathbb{E}[X_{(k)}] = \frac{k}{n+1}.$$

So the expected value of the  $k$ th smallest of  $n$  uniforms is at rank  $k/(n+1)$  between 0 and 1. In particular, the sample median of an odd sample,  $X_{((n+1)/2)}$ , has expectation  $(n+1)/(2(n+1)) = 1/2$  in the uniform case. More generally, sample quantiles are order statistics, and their distributions can be analyzed via these formulas; this leads to confidence intervals for quantiles.

**Example 42** (Maximum of standard Gaussians and the  $\sqrt{2 \log n}$  scale). Let  $Z_1, \dots, Z_n$  be i.i.d.  $N(0, 1)$  and denote

$$M_n := \max_{1 \leq i \leq n} Z_i.$$

A classical question is: how large is  $M_n$  typically, as a function of  $n$ ?

*Step 1: distribution of the maximum.* For any real  $t$ ,

$$\mathbb{P}(M_n \leq t) = \mathbb{P}(Z_1 \leq t, \dots, Z_n \leq t) = \prod_{i=1}^n \mathbb{P}(Z_i \leq t) = \Phi(t)^n,$$

where  $\Phi$  is the standard normal cdf. Thus

$$\mathbb{P}(M_n > t) = 1 - \Phi(t)^n.$$

*Step 2: an upper tail bound for a single Gaussian (Mills ratio).* Let  $\varphi(t) = \frac{1}{\sqrt{2\pi}} e^{-t^2/2}$  be the standard normal density. For  $t > 0$ ,

$$1 - \Phi(t) = \int_t^\infty \varphi(u) du.$$

Since  $u \geq t$  for  $u \geq t$ ,

$$1 - \Phi(t) = \int_t^\infty \varphi(u) du \leq \int_t^\infty \frac{u}{t} \varphi(u) du = \frac{1}{t} \int_t^\infty u \varphi(u) du.$$

But  $\varphi'(u) = -u\varphi(u)$ , so

$$\int_t^\infty u \varphi(u) du = [-\varphi(u)]_{u=t}^\infty = \varphi(t).$$

Hence the (one-sided) bound:

$$1 - \Phi(t) \leq \frac{\varphi(t)}{t}, \quad t > 0.$$

*Step 3: bounding  $\mathbb{E}[M_n]$  via the positive part and tails.* The maximum  $M_n$  can be negative (if all  $Z_i$  are negative), so the standard tail formula  $\mathbb{E}[X] = \int_0^\infty \mathbb{P}(X > t) dt$  does not apply directly. Instead, consider the nonnegative *positive part*

$$M_n^+ := \max(M_n, 0) \geq M_n.$$



Then

$$\mathbb{E}[M_n] \leq \mathbb{E}[M_n^+].$$

For a nonnegative r.v.  $W$  with density  $f_W$  one can check, by integrating  $\mathbb{E}[W] = \int_0^\infty t f_W(t) dt$  by parts, that

$$\mathbb{E}[W] = \int_0^\infty \mathbb{P}(W > t) dt.$$

Applying this to  $W = M_n^+$  gives

$$\mathbb{E}[M_n^+] = \int_0^\infty \mathbb{P}(M_n > t) dt,$$

hence

$$\mathbb{E}[M_n] \leq \int_0^\infty \mathbb{P}(M_n > t) dt.$$

*Step 4: using independence + Mills to bound the integral.* For  $0 \leq u \leq 1$  we have  $1 - u^n \leq n(1 - u)$ , so

$$\mathbb{P}(M_n > t) = 1 - \Phi(t)^n \leq n(1 - \Phi(t)), \quad t \geq 0.$$

Therefore, for any  $a > 0$ ,

$$\mathbb{E}[M_n] \leq \int_0^a \mathbb{P}(M_n > t) dt + \int_a^\infty \mathbb{P}(M_n > t) dt \leq a + n \int_a^\infty (1 - \Phi(t)) dt.$$

Using the Mills bound and a simple comparison,

$$\int_a^\infty (1 - \Phi(t)) dt \leq \int_a^\infty \frac{\varphi(t)}{t} dt \leq \frac{1}{a} \int_a^\infty \varphi(t) dt = \frac{1}{a} (1 - \Phi(a)) \leq \frac{\varphi(a)}{a^2},$$

where the last inequality uses our bound again at  $t = a$ . Thus

$$\mathbb{E}[M_n] \leq a + n \frac{\varphi(a)}{a^2}.$$

*Step 5: choosing  $a = \sqrt{2 \log n}$ .* Set

$$a = \sqrt{2 \log n}.$$

Then

$$\varphi(a) = \frac{1}{\sqrt{2\pi}} e^{-a^2/2} = \frac{1}{\sqrt{2\pi}} e^{-\log n} = \frac{1}{\sqrt{2\pi} n},$$

so

$$n \frac{\varphi(a)}{a^2} = n \cdot \frac{1}{\sqrt{2\pi} n} \cdot \frac{1}{a^2} = \frac{1}{\sqrt{2\pi} a^2} = \frac{1}{\sqrt{2\pi} 2 \log n}.$$

We arrive at the explicit bound

$$\boxed{\mathbb{E}[M_n] \leq \sqrt{2 \log n} + \frac{1}{2\sqrt{2\pi} \log n}.$$

## Lectures 24 and 25: Discrete-time Markov chains (intro)

In this part of the course we move from independent samples to *stochastic processes*: sequences of random variables  $(X_0, X_1, X_2, \dots)$  evolving in time. We start with the simplest and most classical case: discrete time, finite or countable state space, and the Markov (memoryless) property.

### Definition and Markov property

Let  $S$  be a finite or countable set, called the *state space*. A stochastic process  $(X_n)_{n \geq 0}$  with values in  $S$  is a *discrete-time Markov chain* if for all integers  $0 \leq m_0 < \dots < m < n$  and all states  $i_0, \dots, i_{n-1}, j \in S$ ,

$$\mathbb{P}\{X_n = j \mid X_m = i, X_{m_{n-2}} = i_{n-2}, \dots, X_{m_0} = i_0\} = \mathbb{P}\{X_n = j \mid X_m = i\}.$$

In words: *given the present state, the future is conditionally independent of the past*. The current state  $X_m$  contains all the information about the past that matters for predicting  $X_n$ .

We focus on the common and very useful case where the transition probabilities do not depend on time.

**Time-homogeneous Markov chain.** A Markov chain is *(time) homogeneous* if

$$\mathbb{P}\{X_{n+1} = j \mid X_n = i\} = p_{ij} \quad \text{does not depend on } n.$$

The constants  $p_{ij}$  are called the *one-step transition probabilities*.

For each state  $i \in S$ ,

$$p_{ij} \geq 0, \quad \sum_{j \in S} p_{ij} = 1.$$

### Transition matrix and $n$ -step transitions

Collect the transition probabilities into a matrix

$$P = (p_{ij})_{i,j \in S},$$

called the *transition matrix*. Each row of  $P$  is a probability distribution (nonnegative entries summing to 1); such matrices are often called *stochastic matrices*.

For  $n \geq 0$  define the  $n$ -step transition probabilities

$$p_{ij}^{(n)} := \mathbb{P}\{X_n = j \mid X_0 = i\}, \quad i, j \in S.$$

The Markov property plus conditioning on an intermediate time gives the *Chapman-Kolmogorov equations*

$$p_{ij}^{(n+m)} = \sum_{k \in S} p_{ik}^{(n)} p_{kj}^{(m)}, \quad n, m \geq 0.$$

C-K equation 的重要结论:

In matrix form, if we denote  $P^{(n)} = (p_{ij}^{(n)})$ , this says

$$P^{(n+m)} = P^{(n)} P^{(m)}.$$

With the initial condition  $P^{(0)} = I$  (identity matrix), it follows by induction that

$$P^{(n)} = P^n, \quad n = 0, 1, 2, \dots$$

So the  $n$ -step transition probabilities are just the  $n$ th matrix power of  $P$ :

$$p_{ij}^{(n)} = (P^n)_{ij}.$$

## Distribution of the chain at time $n$

It is convenient to record the full distribution of the chain at time  $n$  as a row vector

$$\pi^{(n)} = (\pi_j^{(n)}, j \in S), \quad \pi_j^{(n)} := \mathbb{P}\{X_n = j\}.$$

We think of  $\pi^{(n)}$  as the *distribution of the chain at time  $n$* .

Using the law of total probability and the one-step transition matrix,

$$\pi_j^{(n+1)} = \sum_{i \in S} \mathbb{P}\{X_n = i\} \mathbb{P}\{X_{n+1} = j \mid X_n = i\} = \sum_{i \in S} \pi_i^{(n)} p_{ij},$$

or in matrix notation

$$\pi^{(n+1)} = \pi^{(n)} P.$$

Iterating,

$$\pi^{(n)} = \pi^{(0)} P^n, \quad n = 0, 1, 2, \dots$$

**Lemma (matrix rules for homogeneous chains).** For every homogeneous Markov chain with transition matrix  $P$ ,

(对先前结论的总结)

1.  $P^{(n+m)} = P^{(n)} P^{(m)}$  for all  $n, m \geq 0$ ;
2.  $P^{(n)} = P^n$  (ordinary matrix powers);
3.  $\pi^{(n+1)} = \pi^{(n)} P$ ;
4.  $\pi^{(n)} = \pi^{(0)} P^n$ .

Thus the behaviour of the chain is completely determined by the initial distribution  $\pi^{(0)}$  and the one-step matrix  $P$ .

## Examples of finite Markov chains

**Example 43** (Two-state “machine” model). Suppose a machine can be either

$$0 = \text{working}, \quad 1 = \text{broken}.$$

Each day, a working machine breaks with probability  $a \in (0, 1)$ , and a broken machine is repaired with probability  $b \in (0, 1)$ . Then

$$P = \begin{pmatrix} 1-a & a \\ b & 1-b \end{pmatrix}.$$

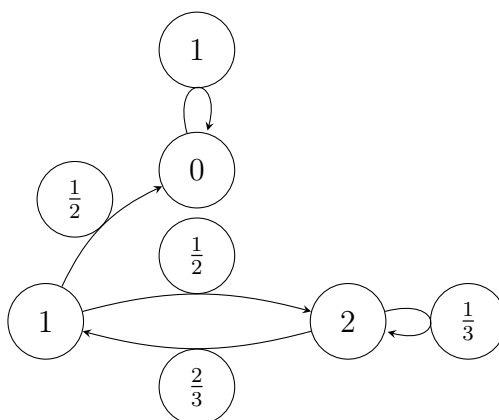
For instance, starting from state 0 (working), the probability it is broken after two days is

$$p_{01}^{(2)} = (P^2)_{01} = (1-a)a + a(1-b).$$

**Example 44** (Three-state chain and its graph). Let the state space be  $S = \{0, 1, 2\}$  and

$$P = \begin{pmatrix} 1 & 0 & 0 \\ 1/2 & 0 & 1/2 \\ 0 & 2/3 & 1/3 \end{pmatrix}.$$

Row  $i$  tells us where we can go from state  $i$  in one step and with what probabilities. It is often convenient to visualize such a chain as a directed graph:



From this picture it is easy to read off qualitative features: state 0 is absorbing, 1 and 2 form a class that eventually gets absorbed at 0, etc.

**Example 45** (Toy weather model). Let the state space be

$$S = \{\text{sunny } (S), \text{ cloudy } (C), \text{ rain } (R)\}.$$

A simple model could be

$$P = \begin{pmatrix} 0.7 & 0.2 & 0.1 \\ 0.3 & 0.4 & 0.3 \\ 0.2 & 0.5 & 0.3 \end{pmatrix}.$$

A typical question would be: “If today is cloudy, what is the probability that two days from now it is raining?” Answer: compute  $P^2$  and look at the entry in row  $C$ , column  $R$ .

## Communicating classes, essential and transient states

For qualitative questions (“Where can the chain go?” “Does it come back?”) we need a bit of terminology. For the rest of this lecture we keep the state space finite.

**Accessibility and communication.** We say that *state  $j$  is accessible from  $i$*  if for some  $n \geq 0$

$$p_{ij}^{(n)} = \mathbb{P}\{X_n = j \mid X_0 = i\} > 0.$$

If  $j$  is accessible from  $i$  and  $i$  is accessible from  $j$ , we say that  $i$  and  $j$  *communicate* and write  $i \leftrightarrow j$ . It is easy to check that  $\leftrightarrow$  is an equivalence relation; its equivalence classes are called *communicating classes*.

**Essential vs. inessential states.** A state  $i$  is called *inessential* if there exists some state  $j$  and some  $N > 0$  such that  $p_{ij}^{(N)} > 0$  but  $p_{ji}^{(M)} = 0$  for all  $M > 0$ : the chain can move from  $i$  to  $j$  with positive probability but can never return from  $j$  to  $i$ . Intuitively, once the chain escapes from  $i$  it may be gone forever.

对 finite MC  
inessential  
 $\Leftrightarrow$  transient

A state that is not inessential is called *essential*. In a finite chain this is equivalent to saying that  $i$  belongs to a closed communicating class (there are no transitions from that class to the outside).

**Absorbing states.** A state  $i$  is called *absorbing* if  $p_{ii} = 1$ . Once the chain enters  $i$ , it stays there forever. A class consisting of a single absorbing state is called an *absorbing class*.

**Example 46** (Small graph classification). Consider again

$$P = \begin{pmatrix} 1 & 0 & 0 \\ 1/2 & 0 & 1/2 \\ 0 & 2/3 & 1/3 \end{pmatrix}.$$

State 0 is absorbing (row 0 is  $(1, 0, 0)$ ). States 1 and 2 communicate with each other (you can go back and forth between them), but with positive probability you eventually hit 0 and then stay there, so  $\{1, 2\}$  is a transient (inessential) class.

## Recurrence and transience (via returns)

Fix a state  $i \in S$ . We are interested in the behaviour of the chain when it *starts* from  $i$ . To keep track of this, it is convenient to use a special notation:

- $\mathbb{P}_i(\cdot)$  means probability under the assumption that  $X_0 = i$ :

$$\mathbb{P}_i(A) := \mathbb{P}(A \mid X_0 = i).$$

Fix  $i \in S$  and define the *first return time* to  $i$  by

$$T_i := \inf\{n \geq 1 : X_n = i\},$$

i.e. the first time  $n \geq 1$  at which the chain comes back to state  $i$  (after time 0). We say:

- $i$  is **recurrent** if  $\mathbb{P}_i(T_i < \infty) = 1$ ; the chain returns to state  $i$  with probability one at some finite time;
- $i$  is **transient** if  $\mathbb{P}_i(T_i < \infty) < 1$ ; there is a positive probability that, once the chain leaves  $i$ , it will never come back.

Intuitively, from a recurrent state you keep coming back infinitely often; from a transient state there is a positive chance you eventually leave forever. In a finite chain, every state in a closed communicating class is recurrent, and every inessential state is transient.

**Theorem (number of returns).** For a homogeneous discrete Markov chain and any state  $i$ :

- if  $i$  is recurrent, then starting from  $X_0 = i$  the chain returns to  $i$  infinitely many times with probability 1;
- if  $i$  is transient, then starting from  $X_0 = i$  the chain visits  $i$  only finitely many times with probability 1.

So recurrence/transience tells us whether the chain keeps coming back to a state or not.

## Stationary distributions

A key long-run quantity of interest is the limiting distribution of  $X_n$  as  $n \rightarrow \infty$ . A very important object is the stationary distribution.

**Stationary distribution.** A probability vector  $\pi = (\pi_i)_{i \in S}$  is called *stationary* for a Markov chain with transition matrix  $P$  if

$$\pi \geq 0, \quad \sum_{i \in S} \pi_i = 1, \quad \text{and} \quad \pi = \pi P.$$

Equivalently, for each state  $j$ ,

$$\pi_j = \sum_{i \in S} \pi_i p_{ij}.$$

If  $X_0 \sim \pi$ , then  $\pi^{(n)} = \pi$  for all  $n$ : the distribution is unchanged by the evolution of the chain.

**Example 47** (Stationary distribution for the two-state machine). Recall

$$P = \begin{pmatrix} 1-a & a \\ b & 1-b \end{pmatrix}, \quad a, b \in (0, 1).$$

We look for  $\pi = (\pi_0, \pi_1)$  such that  $\pi = \pi P$  and  $\pi_0 + \pi_1 = 1$ . Writing out  $\pi = \pi P$ ,

$$\begin{cases} \pi_0 = \pi_0(1 - a) + \pi_1 b, \\ \pi_1 = \pi_0 a + \pi_1(1 - b). \end{cases}$$

The two equations are dependent; using, say, the first plus  $\pi_0 + \pi_1 = 1$  gives

$$\pi_0 a = \pi_1 b, \quad \pi_0 + \pi_1 = 1.$$

Thus

$$\pi_0 = \frac{b}{a + b}, \quad \pi_1 = \frac{a}{a + b}.$$

Interpretation: in the long run, the fraction of time the machine is working vs. broken is proportional to the repair/break rates  $b : a$ .

**Example 48** (Random walk on a 3-cycle). Let  $S = \{1, 2, 3\}$  and at each step move clockwise or counterclockwise with probability  $1/2$  each:

$$P = \begin{pmatrix} 0 & 1/2 & 1/2 \\ 1/2 & 0 & 1/2 \\ 1/2 & 1/2 & 0 \end{pmatrix}.$$

By symmetry it is natural to guess the uniform distribution  $\pi = (1/3, 1/3, 1/3)$ . Indeed,

$$\pi P = \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right) P = \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right) = \pi.$$

So the chain spends, in the long run, about one-third of the time in each state.

**Example 49** (Absorbing chain). If a chain has an absorbing state  $i^*$  (with  $p_{i^*i^*} = 1$ ), then the probability vector concentrated at  $i^*$ ,

$$\pi_j = \begin{cases} 1, & j = i^*, \\ 0, & j \neq i^*, \end{cases}$$

satisfies  $\pi = \pi P$  and is therefore stationary. This reflects the fact that once absorbed, the chain never leaves  $i^*$ .

## Additional Reading: Ergodic chains and limiting distributions

Ultimately we care about the behaviour of  $P^n$  for large  $n$ , i.e. about the limit of  $\pi^{(0)} P^n$  as  $n \rightarrow \infty$ . In many problems one simply assumes that a limit exists; this section explains what that means mathematically.

**Definition 17** (Limiting distribution and ergodicity). A probability vector  $\pi^* = (\pi_j^*)_{j \in S}$  is called a *limiting distribution* for a Markov chain if for every starting state  $i \in S$ ,

$$\lim_{n \rightarrow \infty} p_{ij}^{(n)} = \pi_j^* > 0 \quad \text{for all } j \in S.$$

Equivalently, for every initial distribution  $\pi^{(0)}$ ,

$$\pi^{(0)} P^n \longrightarrow \pi^* \quad \text{as } n \rightarrow \infty.$$

A (finite, homogeneous) Markov chain that admits such a limiting distribution is called *ergodic*.

The limiting distribution (if it exists) is automatically stationary: taking limits in  $\pi^{(0)} P^{n+1} = \pi^{(0)} P^n P$  shows that  $\pi^* = \pi^* P$ . So in nice cases the long-run behaviour can be found just by solving the linear system  $\pi = \pi P$ . To state a clean theorem we need one more definition.

**Definition 18** (Irreducibility and period). A chain is *irreducible* if it has a single communicating class (every state is accessible from every other state).

For a state  $i$  define its *period*

$$d(i) := \gcd\{n \geq 1 : p_{ii}^{(n)} > 0\},$$

where gcd is the greatest common divisor. If  $d(i) = 1$  the state is called *aperiodic*. In an irreducible chain all states have the same period, so we can speak of the period of the chain. An irreducible chain is called *aperiodic* if some (and hence every) state has period 1.

**Example 50** (A simple periodic chain). Consider the two-state chain on  $S = \{0, 1\}$  with transition matrix

$$P = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

The chain deterministically flips between 0 and 1:

$$0 \rightarrow 1 \rightarrow 0 \rightarrow 1 \rightarrow \dots$$

Starting from 0 we have

$$p_{00}^{(1)} = 0, \quad p_{00}^{(2)} = 1, \quad p_{00}^{(3)} = 0, \quad p_{00}^{(4)} = 1, \quad \dots$$

so returns to 0 occur only at even times. The period of state 0 is

$$d(0) = \gcd\{n \geq 1 : p_{00}^{(n)} > 0\} = \gcd\{2, 4, 6, \dots\} = 2,$$

and similarly  $d(1) = 2$ . This is a *periodic* Markov chain of period 2.



**Ergodic theorem for finite chains.** Let  $P$  be the transition matrix of a finite homogeneous Markov chain. The following are equivalent:

- the chain is irreducible and aperiodic;
- the chain is ergodic in the sense above.

In this case:

1. there exists a unique stationary distribution  $\pi$  with  $\pi = \pi P$ ;
2. for every initial distribution  $\pi^{(0)}$ ,

$$\pi^{(0)} P^n \longrightarrow \pi \quad \text{as } n \rightarrow \infty;$$

equivalently,  $\lim_{n \rightarrow \infty} p_{ij}^{(n)} = \pi_j$  for all states  $i, j$ .